PEARSON EDEXCEL INTERNATIONAL A LEVEL BIOLOGY · WBI11

Unit 1: Molecules, Diet, Transport and Health

Revision notes, spec point by spec point (Topics 1 and 2)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic 1 – Molecules, Transport and Health

SPEC 1.1 Water: a polar solvent for transport

WHAT THIS SPEC POINT ASKS

Explain why water's dipole nature makes it a good solvent, and why that matters for transport.

1.1 Why water is a dipole

- In a water molecule, oxygen is more **electronegative** than hydrogen, so it pulls the shared electrons in each O–H covalent bond towards itself.
- Oxygen ends up slightly negative (**δ−**) and each hydrogen slightly positive (**δ+**). The molecule is bent, so the charges don't cancel: water is **polar** (a **dipole**).
- The δ+ hydrogen of one molecule is attracted to the δ− oxygen of a neighbour. This weak attraction is a **hydrogen bond**. Hydrogen bonds constantly break and re-form, which is why water is a liquid that flows.

1.1 Why that makes water a good solvent

- Ions and polar molecules are attracted to water's charged regions:
  - δ− oxygens cluster around **positive ions** (e.g. Na^+); δ+ hydrogens cluster around **negative ions** (e.g. Cl^-).
  - The ions are pulled apart from each other, surrounded by water molecules, and **dissolve**.
- Polar molecules such as **glucose, amino acids** and **urea** form hydrogen bonds with water, so they dissolve too.
- Non-polar molecules (e.g. **lipids**) can't interact with water's charges, so they don't dissolve. Blood carries them another way (e.g. packaged as lipoproteins).
- Once dissolved, solutes can **react** (water is the medium for metabolic reactions) and can be **carried in solution**.

1.1 Link to transport

Where	Carried in solution
Blood plasma	glucose, amino acids, ions, hormones, urea, CO_2 (mostly as hydrogencarbonate)
Xylem (plants)	water and mineral ions
Phloem (plants)	sucrose and amino acids

MARK-SCHEME LANGUAGE

- "oxygen (atom) is slightly negative and the hydrogens (atoms) are slightly positive". The charges must be **partial** (δ+ / δ−, "slight"), never full + and −.
- Why: "uneven charge distribution" / "oxygen pulls the electrons towards it".
- Diagram: two H joined to one O by covalent bonds, δ− on the O, δ+ on each H. A hydrogen bond runs from the δ+ H of one molecule to the δ− O of the next.
- Ions dissolve because water is "polar / dipolar": "sodium ions attracted to the oxygen", "chloride ions attracted to the hydrogens".
- Fatty acids are insoluble: "non-polar / hydrophobic tails" that "do not form hydrogen bonds" with water. "Too big" is ignored.
- "dipolar" is the preferred cloze word; "polar" is accepted.
- Solutes are "transported in solution" in plasma, xylem and phloem; water is the "medium for (metabolic) reactions".

Often asked as: draw a water molecule with partial charges, draw a hydrogen bond, draw water around Na^+, explain why something dissolves (or doesn't), describe a solubility-against-temperature graph.

COMMON MISTAKE

Saying water forms hydrogen bonds *with ions*. Ions are **attracted to the charged poles** of water molecules. Hydrogen bonds form between water and **polar molecules**, or between water molecules. And the oxygen and hydrogens are **atoms**, not "molecules": that slip loses the mark.
Full + and − charges, or charges on the wrong atoms. Hydrogen bonds drawn H to H or O to O.
"Glucose is too large to dissolve." It dissolves well because it is polar.
Calling a gradient a "rate" when time isn't on the x-axis.

BEYOND THE SPEC, BUT EXAMINED

Cohesion (water molecules hydrogen-bond to each other, so a column of water can be pulled up the xylem), **adhesion** (water sticks to vessel walls) and **high specific heat capacity** (blood and cells resist sudden temperature changes). Explain each one through hydrogen bonding.
Glucose is very soluble because its many **hydroxyl (–OH) groups** hydrogen-bond with water (WBI11 Jan 2020).
Mr of glucose: $C_6H_{12}O_6$ = (6 × 12) + (12 × 1) + (6 × 16) = **180**. You need to know the formula (WBI11 Jan 2020).

SPEC 1.2

Mono-, di- and polysaccharides

WHAT THIS SPEC POINT ASKS

(i) Know how monosaccharides, disaccharides and polysaccharides differ, including glycogen and starch (amylose and amylopectin). (ii) Relate their structures to their roles in providing and storing energy. (β-glucose and cellulose come later, in Topic 4.)

1.2 The three sizes of carbohydrate

Type	What it is	Examples	Soluble? Sweet?
Monosaccharide	single sugar unit (monomer), $(CH_2O)_n$	glucose, fructose, galactose (hexoses, $C_6H_{12}O_6$); ribose, deoxyribose (pentoses)	yes, yes
Disaccharide	two monosaccharides joined by a **glycosidic bond**	maltose, sucrose, lactose	yes, yes
Polysaccharide	many monosaccharides joined in a chain (polymer)	starch (amylose + amylopectin), glycogen	no, no

1.2 α-glucose

- Glucose is a **hexose** (6 carbons). Five carbons and an oxygen form the ring; carbon 6 sticks out as CH_2OH.
- In **α-glucose** the –OH on **carbon 1 points down**, on the same side as the –OH on carbon 4. (β-glucose, with the C1 –OH up, is Topic 4.)
- Glucose is the main **respiratory substrate**: small and **soluble**, so it is easily transported in blood and taken into cells, and its bonds store energy that respiration releases to make ATP.
- Disaccharides are quick energy too: they are digested into monosaccharides.

1.2 Storage polysaccharides: structure → function

	Amylose (starch)	Amylopectin (starch)	Glycogen (animals)
Glycosidic bonds	**1,4** only	1,4 and **1,6**	1,4 and **1,6**
Shape	unbranched chain that coils into a **helix**	**branched** (about every 20–30 glucose units)	**highly branched** (about every 8–12 units), more compact
What that gives	**compact**: lots of glucose in a small space	many **ends** → fast hydrolysis	most ends → **fastest** glucose release, suits animals' high metabolic rate
Where	starch grains in plant cells (amyloplasts/chloroplasts)		granules in **liver** and **muscle** cells

Properties that make all three good stores, with the reason for each:

- **Insoluble** → doesn't lower the **water potential** of the cell, so water isn't drawn in by osmosis; it also can't diffuse out of the cell.
- **Large** → can't cross the cell membrane, so the store stays where it is needed.
- **Compact** (coiled or branched) → a lot of energy stored in a small volume.
- **Branched** (amylopectin, glycogen) → many terminal glucose units, so many enzyme molecules can hydrolyse it at the same time, releasing glucose rapidly.
- Hydrolysis gives **α-glucose**, which can be respired straight away.

EXAM TIP

"Relate structure to function" questions want **pairs**: feature → so → consequence. "Branched, *so* there are many ends for enzymes to act on, *so* glucose is released quickly for respiration."

MARK-SCHEME LANGUAGE

- Glycogen: "polysaccharide / polymer of α-glucose", "branched", "1–4 and 1–6 glycosidic bonds". Not "made of amylose and amylopectin".
- Link each feature to storage: "large, therefore insoluble / no osmotic effect" (and it can't diffuse out); "branched, so hydrolysis is faster"; "compact, so high energy density". "Easier to break down" is ignored.
- Amylose v amylopectin: **both** made of α-glucose, **both** have 1–4 bonds, both contain C, H and O only. Amylose: 1–4 only, unbranched, helical. Amylopectin: 1–4 and 1–6, branched.
- Starch is **insoluble**.

Often asked as: tick-grids (amylose v amylopectin v glycogen), compare-and-contrast, describe glycogen's structure, explain how glycogen suits storage, 6-mark on carbohydrate digestion, transport and storage.

COMMON MISTAKE

Stating "insoluble" on its own. Link it: large and insoluble, **so it has no osmotic effect** (it doesn't lower the cell's water potential) and can't diffuse out.
Thinking amylopectin has no 1–4 bonds: it has both 1–4 and 1–6.
Compare questions: give the **similarities** too, as paired statements ("both…", "amylose… but amylopectin…"). Points from two separate descriptions aren't pieced together.
Writing about properties when the question asks about structure. Glycogen isn't in plants, and starch isn't in animals.

BEYOND THE SPEC, BUT EXAMINED

- Monosaccharide general formula: $\mathbf{C_nH_{2n}O_n}$ (WBI11 Jan 2023).
- Liver glycogen is hydrolysed when blood glucose is too low; excess glucose is turned into fat (WBI11 Oct 2025).
- Undigested starch passes out in the faeces, so less glucose is absorbed. Starch itself is never absorbed (WBI11 Jan 2024).

SPEC 1.3

Core Practical 1: sugars and starch by colour standards

WHAT THIS SPEC POINT ASKS

Use Benedict's reagent and iodine solution, with colour standards, to estimate (semi-quantitatively) the concentration of reducing sugar and of starch.

1.3 Why Benedict's works

- A **reducing sugar** (all monosaccharides; maltose and lactose) can donate electrons. It reduces the blue **copper(II) sulfate** in Benedict's reagent to **copper(I) oxide**, a **brick-red precipitate**.
- More reducing sugar → more precipitate → colour runs **blue** → **green** → **yellow** → **orange** → **brick-red**.

CORE PRACTICAL 1: METHOD IN BRIEF

Reducing sugar (Benedict's)

1. Make a **dilution series** of glucose of known concentration (e.g. 0, 0.1, 0.2, 0.5, 1.0 mol dm^{-3}). For a **serial dilution**, dilute by the same factor each step (e.g. halve each time).

2. Add the **same volume** of Benedict's reagent (in **excess**) to the **same volume** of each solution.
3. Heat in a **water bath at the same temperature** (mark schemes use **80–95 °C**; some lab sheets say 60–80 °C, and either works if it is kept constant) for the **same time** (e.g. 5 minutes).
4. These tubes are your **colour standards**.
5. Test the unknown in exactly the same way and match its colour to the nearest standard.

Starch (iodine): make a starch dilution series and add the same volume of iodine solution to each. Colour runs **orange-brown** → **blue-black** as starch concentration rises. Compare the unknown with the standards. (No heating.)

1.3 Making it more accurate and reliable

- **Control variables**: volumes of sample and reagent, temperature, heating time, time before the colour is judged.
- Matching colours by eye is **subjective**, so the method is only *semi*-quantitative. Improve it with a **colorimeter**:
  1. Zero it with a blank (water), using a filter of contrasting colour.
  2. Measure the absorbance (or transmission) of each standard.
  3. Plot a **calibration curve** of absorbance against known concentration.
  4. Measure the unknown and **read its concentration off the curve**.
- Use **smaller concentration intervals** between standards for a more precise estimate.
- **Repeat** and calculate a mean; view the tubes against a **white tile/background**.

BEYOND THE SPEC, BUT EXAMINED

Non-reducing sugars (e.g. sucrose) give a negative Benedict's result. To test for them: boil with **dilute hydrochloric acid** (hydrolyses the glycosidic bond), **neutralise** with sodium hydrogencarbonate, then repeat the Benedict's test. A colour change now shows a non-reducing sugar was present.
Unfamiliar colorimetric tests: use what each disaccharide is made of to predict results (e.g. a test for fructose detects sucrose but not maltose or lactose). If the calibration line doesn't pass through the origin, the test isn't sensitive enough to detect very low concentrations (WBI11 Jun 2023).

MARK-SCHEME LANGUAGE

- "same volume of Benedict's", "same temperature and time of heating", "excess reagent", "compare with colour standards", "calibration curve", "colorimeter (removes subjectivity)".
- Calibration line not through the origin: "test is not sensitive enough" below a threshold concentration.

Often asked as: rarely examined step by step in WBI11 (2019–2026), but learn the method. More often: an unfamiliar colorimetric assay, or Benedict's as an MCQ distractor.

COMMON MISTAKE

Proposing Benedict's to detect sucrose without hydrolysing it first. Sucrose is a **non-reducing** sugar.

SPEC 1.4 Condensation and hydrolysis: glycosidic bonds

WHAT THIS SPEC POINT ASKS

Know how monosaccharides join by condensation reactions, forming glycosidic bonds, to make maltose, sucrose, lactose and polysaccharides, and how hydrolysis splits them.

1.4 Two mirror-image reactions

- **Condensation**: an –OH group on each sugar reacts; **one molecule of water is removed** and a **glycosidic bond** (C–O–C) forms between them.
- **Hydrolysis** (hydro = water, lysis = splitting): **water is added** across the glycosidic bond, breaking it. In living things it is catalysed by enzymes (amylase, maltase, sucrase, lactase), e.g. in digestion, or when liver and muscle cells break down glycogen.

1.4 Which sugars make which disaccharide

Disaccharide	Made from	Bond
Maltose	glucose + glucose	1,4-glycosidic
Sucrose	glucose + fructose	1,2-glycosidic
Lactose	glucose + galactose	1,4-glycosidic

Memory aid: **M**altose = a **M**atching pair; **S**ucrose has the fruit sugar; **L**actose is in mi**L**k (ga**L**actose).

- The bond is named by the carbons it joins: C1 of one sugar to C4 of the next is a **1,4** bond. Branches (amylopectin, glycogen) are **1,6** bonds.
- Polysaccharides form by **repeated** condensation. A chain of n glucose units releases $n - 1$ molecules of water.

EXAM TIP

To draw the condensation of two α-glucose: show the –OH on C1 of one and on C4 of the other, ring the H_2O being removed, then draw the product with **one O bridging C1 and C4**. Label it "1,4-glycosidic bond" and write **+ H_2O** as a product.

WORKED EXAMPLE: MR OF A DISACCHARIDE

Glucose ($C_6H_{12}O_6$) has Mr 180. Each condensation removes one H_2O (Mr 18).
Maltose: 180 + 180 − 18 = **342** ($C_{12}H_{22}O_{11}$).
A chain of n glucose units: $180n - 18(n - 1)$.

MARK-SCHEME LANGUAGE

- "condensation", "glycosidic (bond)", "hydrolysis" (**not** "hydration").
- Lactose = "galactose and glucose"; sucrose = glucose + fructose; maltose = glucose + glucose.
- Drawing: "glycosidic bond drawn correctly between C1 and C4" and water shown as a product. For hydrolysis, show H and OH added back to C1 and C4.
- Equation: $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + \mathbf{H_2O}$. The water is the part most often left out.
- Enzyme specificity: "the active site (of amylase) is complementary to the 1–4 glycosidic bond".

Often asked as: draw the condensation of two monosaccharides, write the equation, calculate an Mr, name the products of an enzyme acting on a polysaccharide.

COMMON MISTAKE

Swapping condensation and hydrolysis. Writing "lactase" (the enzyme) when you mean "lactose" (the sugar). Pairing the wrong monomers: lactose is glucose + **galactose**, not fructose.
In drawings, forgetting the water or losing atoms: every ring carbon keeps its H and OH groups.

BEYOND THE SPEC, BUT EXAMINED

- Amylase hydrolyses only **1–4** bonds, so digesting amylopectin leaves small branched fragments that still contain the 1–6 bonds (WBI11 Jun 2024).
- Unfamiliar disaccharides appear. Work out the bond name from the carbon numbers drawn (WBI11 Jan 2022).

SPEC 1.5 Triglycerides: ester bonds, saturated vs unsaturated

WHAT THIS SPEC POINT ASKS

(i) Know how a triglyceride is made by condensation between glycerol and three fatty acids, forming ester bonds. (ii) Know how saturated and unsaturated lipids differ.

1.5 Building a triglyceride

- **Glycerol** is an alcohol with **three –OH groups**. A **fatty acid** is a –COOH (carboxyl) group attached to a hydrocarbon chain (written R–COOH).
- The –OH of glycerol reacts with the –COOH of a fatty acid in a **condensation** reaction: one H_2O is removed and an **ester bond** (–O–CO–) forms.
- One glycerol + three fatty acids → one triglyceride + **three water molecules**, with **three ester bonds**.
- **Hydrolysis** (catalysed by **lipase**) reverses this.

1.5 Saturated vs unsaturated

	Saturated	Unsaturated
C=C double bonds in the hydrocarbon chain	**none**: every carbon carries the maximum number of H atoms	**one** (mono-) or **more** (poly-unsaturated)
Chain shape	straight	**kinked** at each C=C
Packing	molecules pack closely	can't pack closely
At room temperature	usually **solid** (fats); mostly animal	usually **liquid** (oils); mostly plant
Health link	raise LDL and blood cholesterol → CVD risk (1.18)	tend to lower LDL

1.5 Why triglycerides make a good energy store

- Many **C–H bonds** → about **twice the energy per gram** of carbohydrate.
- **Insoluble** (non-polar, hydrophobic) → no effect on water potential.
- Other roles: thermal **insulation**, **protection** of organs, buoyancy.

EXAM TIP

Melting-point data questions: melting point **rises with chain length** in saturated fatty acids and **falls as the number of C=C bonds rises**. If you say "positive correlation", **name both variables** or the mark isn't given.

MARK-SCHEME LANGUAGE

- "one glycerol and three fatty acids", "condensation reaction", "ester bond" (esterification accepted), "catalysed by enzymes".
- Unsaturated: "at least one fatty acid has a C=C double bond"; fewer H atoms; the chain is "kinked".
- Saturated: say "no **C=C** double bonds". Plain "no double bonds" is rejected, because the C=O in –COOH is a double bond.
- Compare: **both** contain C, H and O, a hydrocarbon chain and a –COOH group. Ester bonds are not a feature of a fatty acid.
- Triglyceride v phospholipid: both have glycerol, fatty acids and ester bonds; a triglyceride has 3 fatty acids, a phospholipid has 2 plus a phosphate group.

Often asked as: describe how a triglyceride forms, draw one from given shapes (three fatty acids, three bonds, 4 bonds per carbon), compare saturated and unsaturated, interpret melting-point graphs.

COMMON MISTAKE

Calling a triglyceride a polymer: it is glycerol + 3 fatty acids, not a chain of repeating monomers. The bond is an **ester** bond (not glycosidic, not peptide), and **three** water molecules are released, not one. Writing "glyceride + 3 fatty acids" or "3 glycerides". Drawing a saturated chain when an unsaturated one was asked for.

BEYOND THE SPEC, BUT EXAMINED

- **C : H ratio**: a saturated fatty acid has more H, so a **lower** C : H ratio than an unsaturated one of the same length (WBI11 Jan 2026).
- Polyunsaturated fats (most C=C bonds) carry the lowest CVD risk (WBI11 Jun 2019).

SPEC 1.6

Why many animals need a heart and circulation

WHAT THIS SPEC POINT ASKS

Understand why many animals have a heart and circulation: mass transport overcomes the limitations of diffusion in meeting an organism's needs.

1.6 The problem: diffusion is only fast over short distances

All cells need a steady supply of O_2 and glucose, and must get rid of CO_2 and urea. A small organism can rely on diffusion. A large one can't, for two reasons:

1. **Distance**: many cell layers mean long diffusion distances, and the time diffusion takes rises steeply with distance.
2. **Surface area : volume ratio falls as size increases**, so the surface is too small to supply the whole volume. Large, active animals also have a **high metabolic rate** (high demand).

WORKED EXAMPLE: SA : V

A 1 cm cube: SA = 6 cm^2, V = 1 cm^3 → **6 : 1**.
A 3 cm cube: SA = 6 × 9 = 54 cm^2, V = 27 cm^3 → **2 : 1**.
Tripling the length cut the SA:V ratio to a third, so there is far less surface per unit of volume to supply.
Other shapes: **cylinder** (a vessel, a worm) V = $\pi r^2 l$, curved SA = $2\pi rl$; **sphere** V = $\frac{4}{3}\pi r^3$, SA = $4\pi r^2$.
Use the **radius** (half the diameter): using the diameter is the commonest error. Give the ratio as **x : 1**.

1.6 The solution: mass transport

- **Mass transport** (mass flow) is the **bulk** movement of a fluid in one direction, driven by a **pressure difference** made by a pump (the **heart**), inside **vessels**.
- It carries substances quickly between exchange surfaces (lungs, gut) and every tissue, so the final step into or out of each cell is only a short diffusion distance.
- It also **maintains concentration gradients** at exchange surfaces and in tissues, by constantly bringing new blood and carrying away what has diffused in.

1.6 Single vs double circulation

	Single (e.g. fish)	Double (mammals)
Route	heart → gills → body → heart	heart → lungs → heart → body → heart
Passes through heart per circuit	once	**twice**
Pressure to the body	low (pressure is lost in the gill capillaries)	**high**: blood is re-pressurised after the lungs

Advantages of a double circulation: **low pressure to the lungs** (doesn't damage the delicate capillaries; slower flow allows time for gas exchange) and **high pressure to the body** (fast delivery for an active, endothermic animal). The **septum** keeps oxygenated and deoxygenated blood apart.

BEYOND THE SPEC, BUT EXAMINED

Other animals' circulations. Fish: pressure falls after the gill capillaries, so flow to the body is slow; that is enough because fish aren't endothermic and have a **lower metabolic rate**. Amphibians (WBI11 June 2025): a **three-chambered heart** (two atria, one ventricle, no septum), so some mixing of oxygenated and deoxygenated blood. That is enough for a small animal with a **low O_2 demand**, which also exchanges gases through its moist skin. As in mammals, oxygenated blood from the lungs returns to the **left** atrium. Insects (WBI11 Oct 2021): an **open** circulation, with blood in body cavities rather than vessels; O_2 reaches cells through tracheoles, so oxygenated and deoxygenated blood don't need to be kept apart.

MARK-SCHEME LANGUAGE

- Mass transport: "all the components (of the blood) flow together", "down a pressure gradient", "to overcome the limitations of diffusion".
- Diffusion is "too slow" over large distances. Quote data if given (e.g. 0.5 ms over 1 μm, but 50 ms over 10 μm).
- Organisms with no circulation: every cell is "not far from" the environment, so "diffusion can supply" enough; "low metabolic rate / oxygen demand"; large SA:V.
- A big heart in context (e.g. a giraffe): blood "pumped under high pressure" to reach the head "against gravity".

Often asked as: explain in a named organism (giraffe, butterfly, yak, hydra, frog), define mass transport, calculate SA:V.

EXAM TIP

Name **both** reasons: diffusion distance too long **and** SA:V too small for a high demand. Say what is transported and where, e.g. "O_2 and glucose to respiring cells".

COMMON MISTAKE

Writing about SA:V when the question is about **distance** or **pressure** (e.g. a giraffe's heart). Answer the factor the question points to.
Vague "nutrients and waste": name O_2, glucose and CO_2, and say **where** the concentration gradient is.
In a comparison, describe both animals, not just the mammal.

SPEC 1.7

Blood vessels: structure related to function

WHAT THIS SPEC POINT ASKS

Understand how the structures of capillaries, arteries and veins relate to their functions.

Feature	Artery	Vein	Capillary
Job	carry blood **away** from the heart at **high pressure**	return blood **to** the heart at **low pressure**	**exchange** between blood and tissues
Wall	**thick**: lots of **elastic tissue** and **smooth muscle**, collagen outside	**thin**: little elastic tissue or muscle	**one cell thick** (endothelium only)
Lumen	**narrow** relative to wall	**wide**	about the width of one red blood cell
Valves	no (only where arteries leave the heart)	**yes**	no
Pulse	yes	no	no

Memory aid: **A**rteries go **A**way; veins go **in**.

1.7 Why each feature matters

- **Elastic tissue** (artery): **stretches** when the ventricle forces blood in, then **recoils** in diastole. This **smooths out pressure surges** and keeps pressure up between beats.
- **Smooth muscle**: **contracts** or relaxes to narrow or widen the lumen, controlling blood flow to different organs.
- **Collagen**: strength, so the wall withstands high pressure and doesn't overstretch or burst.
- **Smooth, folded endothelium**: reduces friction; folds let the artery stretch.
- **Wide lumen in veins**: less resistance at low pressure. **Valves** stop **backflow**. Contracting **skeletal muscles** squeeze veins and push blood towards the heart.
- **Capillary wall one cell thick** → **short diffusion distance**. **Narrow lumen** → red cells pass in single file, pressed against the wall (short distance) and moving **slowly** (more time for exchange). Huge **branched network** → large surface area, and every cell is close to a capillary. **Gaps between cells** let plasma out to form tissue fluid and let white cells squeeze through.

MARK-SCHEME LANGUAGE

- Elastic fibres "stretch" when pressure is high, "widening the lumen", then "elastic recoil" "maintains the high blood pressure" and smooths the flow. In "why doesn't it burst?" questions, recoil is ignored: use stretch + "collagen for (tensile) strength".
- Endothelium: "smooth / flattened", "to reduce friction". Smooth muscle: contracts to control flow (vasoconstriction).
- Capillaries: one cell thick → short diffusion distance for gases; pores let plasma and white cells out; narrow, so red cells are close to the wall.
- Veins: valves "prevent backflow" because pressure is low.
- Artery definition: "carries blood away from the heart". Not "carries oxygenated blood" (the pulmonary artery doesn't).
- Drawing an artery TS: lumen + three layers, labelled tunica intima (endothelium), tunica media (smooth muscle + elastic fibres), tunica externa (collagen). Descriptions instead of names are fine. Label lines must touch the layer, and a wrong label cancels a right one.
- Velocity graphs: flow "decreases through the arterioles", is "low in the capillaries" (huge total cross-sectional area), "increases in the venules". Explain velocity, not pressure.

Often asked as: draw and label a TS of an artery or vein, explain a named feature, read velocity or pressure graphs, calculate a lumen area or mean arterial pressure.

COMMON MISTAKE

"Arteries have thick walls *to pump blood*." Arteries don't pump; the heart does. The wall **withstands** high pressure and the elastic tissue **recoils** to maintain it. Don't swap the verbs: muscle **contracts**, elastic tissue **stretches and recoils**. Capillary walls are made of cells; they don't have "cell walls".
Writing "walls" when you mean "elastic fibres", or mixing up elastic fibres and smooth muscle.
Vein valves drawn the wrong way round: they open in the direction of flow, towards the heart.
"Glucose diffuses out through the pores": examiners only credit diffusion across the wall for gases. Plasma (carrying glucose) is forced out through the gaps.
Forgetting to multiply by the number of vessels in a surface-area calculation.

BEYOND THE SPEC, BUT EXAMINED

Tissue fluid: at the arterial end of a capillary, **hydrostatic pressure** is higher than the opposing pull of the blood's water potential, so plasma (minus large proteins and cells) is forced out. At the venous end, hydrostatic pressure has fallen, so water moves back in by **osmosis** (plasma proteins keep the blood's water potential low). Excess fluid drains into **lymph** vessels. High blood pressure → more tissue fluid forms (oedema).
Mean arterial pressure (formula given): MAP = diastolic + ⅓(systolic − diastolic). For 120/80 mmHg: 80 + 40 ÷ 3 = **93 mmHg** (WBI11 Jun 2025).
Lumen area: lumen diameter = outer diameter − 2 × wall thickness; halve it for the radius, then πr^2. Convert units first (e.g. mm → μm) (WBI11 Jun 2025).

SPEC 1.8 The cardiac cycle and the heart

WHAT THIS SPEC POINT ASKS

Know the cardiac cycle (atrial systole, ventricular systole and cardiac diastole) and relate the structure and operation of the mammalian heart, including its major blood vessels, to its function. (Myogenic stimulation isn't needed at AS.)

1.8 Heart structure

Diagrams show the heart as if it's facing you, so its **right side is on the left of the page**.

- **Right side**: deoxygenated blood in from the body (**vena cava**) → out to the lungs (**pulmonary artery**).
- **Left side**: oxygenated blood in from the lungs (**pulmonary vein**) → out to the body (**aorta**).
- **Atrioventricular (AV) valves** between atria and ventricles: **tri**cuspid on the **ri**ght, **bi**cuspid (mitral) on the left. **Tendinous cords** stop them turning inside out.
- **Semilunar valves** at the base of the aorta and the pulmonary artery.
- **Left ventricle wall thicker** than the right: it generates a higher pressure to pump blood all round the body. The right only pumps to the nearby lungs, and low pressure there protects the lung capillaries.
- **Atria** have thin walls: they only push blood a short distance, into the ventricles.
- **Septum**: keeps oxygenated and deoxygenated blood separate.
- **Coronary arteries**: supply the heart muscle itself with oxygenated blood.

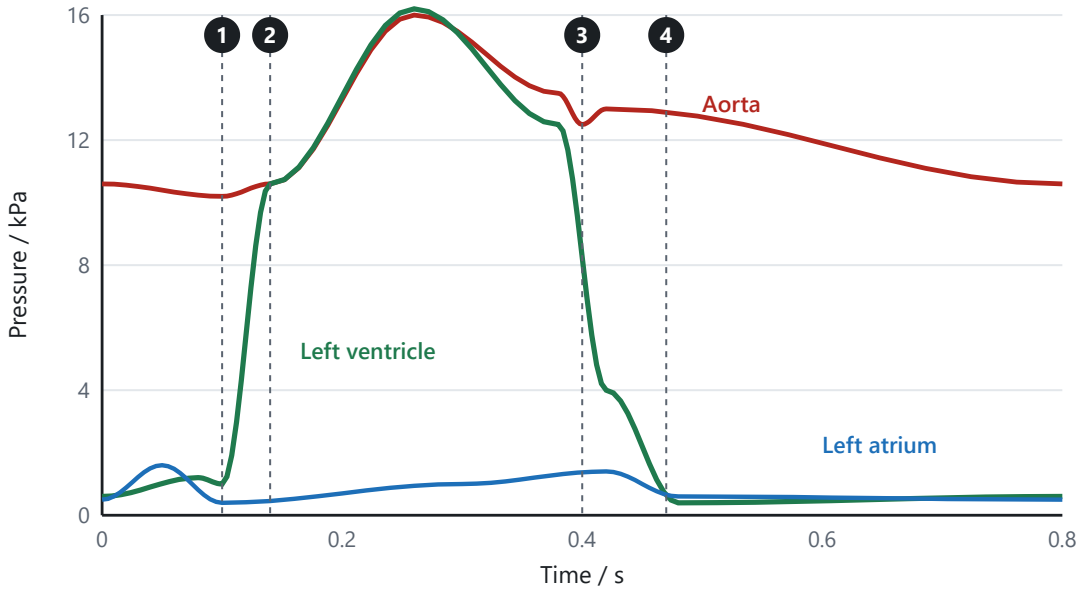
1.8 The three stages (one cycle ≈ 0.8 s at 75 beats per minute)

The pressure rule: **contraction → smaller volume → higher pressure**. Blood always flows from higher to lower pressure.

1. **Atrial systole**: atria contract → atrial pressure rises above ventricular pressure → blood is pushed through the open **AV valves** into the ventricles. Semilunar valves are closed.
2. **Ventricular systole**: ventricles contract (from the apex, at the bottom, upwards) → ventricular pressure rises above atrial pressure → **AV valves shut** (the "lub" sound). When ventricular pressure rises above the pressure in the arteries, the **semilunar valves open** and blood is ejected into the aorta and pulmonary artery.
3. **Cardiac diastole**: atria and ventricles relax → ventricular pressure falls below arterial pressure → **semilunar valves shut** (the "dub" sound). Blood flows from the veins into the atria and, once atrial pressure is above ventricular pressure, the **AV valves open** and the ventricles fill passively.

Stage	Atria	Ventricles	AV valves	Semilunar valves
Atrial systole	contract	relaxed	open	closed
Ventricular systole	relaxed	contract	**closed**	**open**
Diastole	relaxed	relaxed	open	closed

Valves open when pressure behind them is higher than pressure in front, and **close** when that difference reverses. They are moved by pressure; they don't contract.



Left side of the heart. 1: AV valve closes (ventricle > atrium). 2: semilunar valve opens (ventricle > aorta). 3: semilunar valve closes (aorta > ventricle). 4: AV valve opens (atrium > ventricle).

1.8 Reading the pressure graph

- Each point where two lines **cross** is a valve opening or closing.
- From **2** to **3** blood is being ejected into the aorta.
- From **1** to **2** and from **3** to **4**, all valves are shut and the ventricle volume isn't changing.
- The small notch in aortic pressure just after **3** is the semilunar valve snapping shut. Aortic pressure stays high in diastole because the elastic wall **recoils**.
- The right ventricle's curve has the same shape but a much **lower peak** pressure.

WORKED EXAMPLE: HEART RATE FROM A GRAPH

One complete cycle (peak to the *same* point on the next peak) takes 0.8 s.
Beats per second = 1 ÷ 0.8 = 1.25; beats per minute = 1.25 × 60 = **75 bpm**.
(Shortcut: 60 ÷ cycle time in seconds. Reverse: cycle time = 60 ÷ heart rate.)
Follow the s.f. instruction exactly: to 3 s.f. the answer is **75.0**, and "75" loses the mark.

MARK-SCHEME LANGUAGE

- Valve states: atrial systole AV open, SL closed; ventricular systole AV closed, SL open; cardiac diastole AV open, SL closed. The AV valves close at the start of ventricular systole.
- Call it "(cardiac) diastole", not "ventricular diastole" or "diasystole".
- Coronary artery: brings "oxygenated blood to the heart muscle" "for (aerobic) respiration". It branches off the aorta because that's the nearest vessel carrying oxygenated blood. Glucose / nutrients are ignored.
- Exercise: "the cardiac cycle is shorter / completed more frequently", "ventricles contract more forcefully". "Heart rate increases" alone is ignored.
- Right v left ventricle: same timing (both sides "beat simultaneously"); the right peaks lower because it pumps only to the lungs, so the alveoli aren't damaged.
- Delay between atrial and ventricular contraction: so the "atria can finish emptying" into the ventricles before the AV valves close.
- Volume graphs: left ventricle volume increases in atrial systole, decreases in ventricular systole, increases in diastole.

Often asked as: valve-state MCQs and cloze passages, label vessels or draw flow arrows, read pressure or volume graphs, calculate heart rate or cycle length.

BEYOND THE SPEC, BUT EXAMINED

Cardiac output = stroke volume × heart rate (e.g. 70 cm^3 × 75 min^{-1} = 5250 cm^3 min^{-1}). Formally later content, but Unit 1 data questions use it.
Delay at the AV node: the impulse is held up briefly before it passes down the septum, so the atria empty before the ventricles contract. The spec leaves this out at AS, but it has been rewarded (WBI11 Oct 2022).
Pulmonary embolism: a clot lodged in a pulmonary artery stops blood being oxygenated → breathlessness and a raised heart rate (WBI11 Jan 2025).

COMMON MISTAKE

Mixing up pressure and volume, or forgetting to compare. Every valve explanation needs a comparison: "pressure in the ventricle is **higher than** in the atrium, so the AV valve closes."
Swapping the pulmonary artery and pulmonary vein. Thinking the right ventricle contracts after the left: they contract together.
Writing "heart rate increases" when the question asks about events in the cycle. Working out 73 ÷ 60 when you need 60 ÷ 73.

SPEC 1.9 Haemoglobin: carrying O_2 and CO_2

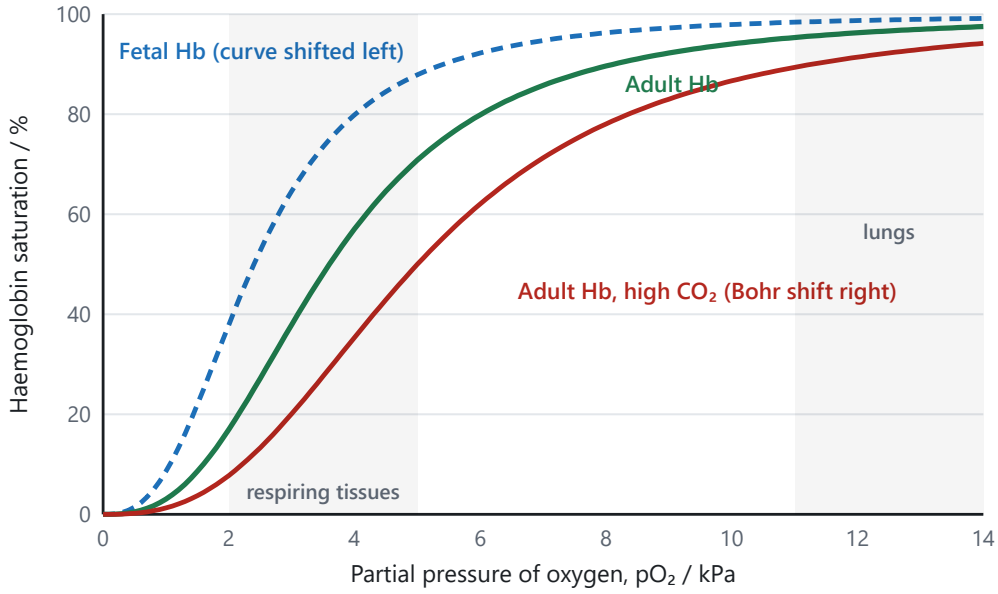
WHAT THIS SPEC POINT ASKS

(i) Understand the role of haemoglobin in transporting oxygen and carbon dioxide. (ii) Understand the oxygen dissociation curve, the Bohr effect, and why fetal haemoglobin's higher affinity for oxygen matters.

1.9 Haemoglobin and oxygen

- Haemoglobin (Hb) is a **globular protein** with **four polypeptide chains** (2 α, 2 β). Each chain holds a **haem group** containing **Fe^{2+}**, which binds one O_2 molecule. So one Hb carries **4 O_2** (8 oxygen atoms).

- Hb + $4O_2$ ⇌ oxyhaemoglobin. **Loading** (association) happens in the lungs; **unloading** (dissociation) in respiring tissues.
- **Cooperative binding**: the first O_2 binds with difficulty. Once it binds, Hb undergoes a **conformational change** (its tertiary/quaternary structure changes), making it easier for further O_2 molecules to bind. This is what makes the curve **S-shaped** (sigmoid).



Shift to the right = lower affinity (O_2 released more readily). Shift to the left = higher affinity (O_2 taken up more readily).

1.9 Reading the curve, in both directions

- **Left → right (loading):** at low pO_2 the curve is shallow: the first O_2 binds with difficulty. In the steep middle, binding is easy after the shape change, so a small rise in pO_2 gives a big rise in saturation. At high pO_2 (lungs) it levels off: Hb is nearly **saturated**, with few free sites.
- **Right → left (unloading):** in the lungs there's little dissociation. In the **steep part**, at the pO_2 of respiring tissues, a small drop in pO_2 releases a **lot** of O_2, exactly where cells need it.
- **Affinity** = how readily Hb takes up O_2. High affinity: loads easily, unloads reluctantly. Low affinity: the opposite.

1.9 The Bohr effect (shift to the right)

1. Respiring tissue produces **CO_2**.
2. In red blood cells, $CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic acid), catalysed by **carbonic anhydrase**. $H_2CO_3 \rightarrow$ **$H^+ + HCO_3^-$**.
3. The H^+ ions (lower pH) bind to Hb and change its shape, **lowering its affinity** for O_2.
4. So **more O_2 is released** at the same pO_2: the curve shifts **right**. Active tissues make the most CO_2, so they get the most O_2.

1.9 Carbon dioxide transport

- About **5%** dissolved in the plasma.
- About **10–20%** bound to Hb as **carbaminohaemoglobin**. The CO_2 joins the **amino groups of the polypeptide chains**, not the haem.
- Most (about **70–85%**) as **hydrogencarbonate ions (HCO_3^-)**: formed in red blood cells as above, then diffuse out into the plasma. Hb picks up the H^+, **buffering** the blood pH. The reaction happens in red cells rather than plasma because that's where the carbonic anhydrase is.

1.9 Fetal haemoglobin (shift to the left)

- Fetal Hb has a **higher affinity** for O_2 than adult Hb.
- At the **placenta** the pO_2 is low. **At the same pO_2**, the mother's Hb unloads O_2 and the fetal Hb, with its higher affinity, **loads** it, so oxygen diffuses from mother to fetus.
- After birth, adult Hb gradually replaces fetal Hb, because the baby now needs Hb that **unloads** O_2 readily in its own respiring tissues.

EXAM TIP

For any curve that's shifted left or right, give three steps: (1) **at the same pO_2**, saturation is higher or lower; (2) so affinity is higher or lower; (3) so O_2 is loaded or unloaded more readily: say *where* and *why that helps*.

MARK-SCHEME LANGUAGE

- Sigmoid curve: "four subunits"; "binding of the first oxygen molecule is difficult"; then "a conformational change" makes "binding of the other molecules easier" (cooperative binding accepted); the curve levels off as Hb becomes saturated. Plain "change of shape" is ignored.
- Structure: 4 polypeptide subunits (2α, 2β), four haem groups, each with an "**iron ion / Fe^{2+}**" (not "iron") that binds "an oxygen molecule / O_2" (not "O"). "Stores oxygen" is rejected. Hydrophilic R groups on the outside make Hb soluble in the red cell's cytoplasm.
- Partial pressure: "the pressure exerted by one gas in a mixture of gases". Not amount or %.
- Fetal Hb: "higher affinity", so "at the same partial pressure" oxygen "leaves the adult haemoglobin and binds to fetal haemoglobin". Write fetus, not "baby".
- Left shift: "higher affinity", "higher % saturation at a lower pO_2", enough O_2 for respiration. Right shift: "lower affinity", so O_2 is unloaded more readily to respiring tissues.
- Altitude: lower pO_2 in the alveoli → smaller gradient → **slower rate of diffusion** → Hb less saturated. "Less diffusion" is ignored.

Often asked as: explain the sigmoid shape, link Hb's structure to its function, explain a shifted curve in an unfamiliar animal (often a 6-mark), read values off a curve.

COMMON MISTAKE

"CO_2 binds to the haem and pushes the oxygen off." It's the **H^+ ions** (from carbonic acid) binding to Hb that lower its affinity. Say "**partial pressure** of oxygen", not "amount of oxygen".
Thinking fetal Hb works because fetal blood has a lower pO_2. At the placenta both bloods are at about the same pO_2: the point is fetal Hb's **higher affinity**.
Writing "iron" for **iron ion (Fe^{2+})**, "speed of dissociation" for **affinity**, or "moves" for **diffuses**. Confusing Hb (the protein) with the red blood cell. Quote values read off the curve.
In multi-animal 6-mark questions, comparing traits across the animals scores; describing one species at a time stays at Level 1.

BEYOND THE SPEC, BUT EXAMINED

Fetal Hb has **γ (gamma) chains** in place of the β chains (WBI11 Oct 2024).
Data questions use other animals. Animals living where O_2 is scarce (high altitude, burrows, mud such as lugworms) have curves **left** of adult human Hb (higher affinity). Small, very active mammals have curves to the **right** (unload readily). **Myoglobin** (one chain, an O_2 store in muscle) has a curve far to the left: it releases O_2 only when pO_2 is very low (WBI11 Jun 2024).

SPEC 1.10

Atherosclerosis

WHAT THIS SPEC POINT ASKS

Understand the sequence of events that leads to atherosclerosis: endothelial dysfunction, inflammatory response, plaque formation, raised blood pressure.

Learn this as a numbered chain. Six-mark answers are marked on the sequence and the links:

1. **Endothelial damage (dysfunction)**: the smooth inner lining of an **artery** is damaged, e.g. by **high blood pressure**, **chemicals in cigarette smoke**, or high blood LDL.
2. **Inflammatory response**: **white blood cells** (macrophages) move into the artery wall at the damaged site.
3. **Cholesterol** (from LDLs in the blood) builds up in the wall and is taken up by the white cells, forming **fatty streaks**.
4. Over time **calcium salts** and **fibrous tissue** build up too → an **atheroma (plaque)** forms. The artery wall hardens and becomes **less elastic**.
5. The plaque **narrows the lumen** → **raised blood pressure** → more damage to the endothelium → more plaque. This is **positive feedback**. A plaque can also **rupture**, which triggers a **blood clot** (1.11) that may block the artery completely.
6. When this happens in a **coronary artery**, less blood, so less **O_2**, reaches the **heart muscle cells** → less **aerobic respiration** → less ATP → the muscle can't contract properly and cells may die (a heart attack, 1.11).

Atherosclerosis happens in **arteries** (high pressure), not in veins or capillaries.

EXAM TIP

Put the steps in order and link each to the next ("this causes…", "leading to…"). Always include **white blood cells / inflammation**, **cholesterol**, **plaque/atheroma** and **narrowing of the lumen**, and finish the chain: **coronary artery** → less O_2 to **heart muscle** → less **aerobic respiration**.

MARK-SCHEME LANGUAGE

- "damage to the endothelial lining" (from high blood pressure, smoking, free radicals) → "inflammatory response" / white blood cells build up → "build-up of cholesterol" in the artery wall → "atheroma / plaque" → "narrowing / blocking" of the **coronary artery** → less "blood / oxygen to the heart muscle cells" for **aerobic respiration**.
- Say "plaque" or "atheroma", not "fatty deposits" or "hardening". Plaque is **not** "caused by blood clotting".
- CVD = any disease of the heart and blood vessels; CHD = disease of the coronary arteries.
- Stents "widen the lumen of the coronary artery", so more blood reaches the heart muscle for respiration.

Often asked as: explain how a factor (smoking, high blood pressure, saturated fat) leads to CVD, 6-mark chains with data, define CVD v CHD.

COMMON MISTAKE

Stopping halfway. Examiners insist on the words **coronary artery** and **heart muscle cells** short of **oxygen** for **aerobic** respiration. Bare "arteries", or "nutrients" instead of O_2, doesn't score.
In "deduce" questions, give conclusions from the data, not a description of it.

BEYOND THE SPEC, BUT EXAMINED

Aneurysm: an artery wall weakened by atheroma bulges and may burst. **Angina**: chest pain when narrowed coronary arteries can't supply enough O_2 to the heart muscle during exertion.
Heart muscle short of O_2: cells switch to anaerobic respiration → lactic acid → lower pH denatures enzymes; much less ATP, so active transport and contraction fail (WBI11 Jan 2021).

SPEC 1.11 Blood clotting

WHAT THIS SPEC POINT ASKS

Understand the blood-clotting process (thromboplastin release, prothrombin → thrombin, fibrinogen → fibrin) and its role in cardiovascular disease (CVD).

1.11 The cascade

1. Damage to a blood vessel (or a ruptured plaque) exposes the tissue beneath; **platelets** stick to it and to each other.
2. Platelets and damaged tissue release **thromboplastin** (an enzyme).
3. Thromboplastin, with **calcium ions (Ca^{2+})**, catalyses the conversion of soluble **prothrombin** into **thrombin** (also an enzyme).
4. Thrombin catalyses the conversion of **soluble fibrinogen** into **insoluble fibrin**.
5. **Fibrin** forms a **mesh** that traps platelets and red blood cells → a **clot**.

Each product activates the next step, so a small trigger is amplified into a large response.

1.11 The link to CVD

- A ruptured atheroma triggers clotting **inside an artery**. The clot (**thrombus**) can block that artery, or a piece can break off and block a narrower vessel further on.
- Blocked **coronary artery** → heart muscle gets less O_2 → less aerobic respiration → less ATP → muscle cells die: a **myocardial infarction** (heart attack).
- Blocked artery to the **brain** → **stroke**.
- **Deep-vein thrombosis (DVT)**: a clot in a deep vein, usually in the leg (risk raised by inactivity, e.g. long flights, and by age).

MARK-SCHEME LANGUAGE

- "platelets release thromboplastin". Platelets are the part most often left out.
- "thromboplastin is an enzyme" that "converts prothrombin into thrombin"; "thrombin is an enzyme" whose active site binds fibrinogen and converts it into fibrin.
- Fibrin forms a "mesh" that "traps platelets / blood cells" → stops bleeding and the entry of bacteria.
- Prothrombin is always present in the plasma as an **inactive precursor**.
- Soluble: thromboplastin, prothrombin, thrombin, fibrinogen. Insoluble: **fibrin** only. Active enzymes: thromboplastin and thrombin.
- Inhibitor questions: say what **won't** happen: "thrombin can't bind to fibrinogen", "less fibrinogen converted into fibrin", "less mesh to trap cells".
- Ca^{2+} and vitamin K are ignored in these answers: fine to mention, but they earn nothing.

Often asked as: cloze or table on the clotting factors, explain the effect of an inhibitor in a new context, 6-mark on DVT leading to pulmonary embolism.

COMMON MISTAKE

Saying fibrinogen turns into "fibres". It becomes **fibrin**, an **insoluble** protein. Remember **thrombin is a soluble enzyme**: it isn't part of the clot. Don't swap the solubility of thrombin and fibrin.
Retelling the whole cascade when the question asks about one step, or describing normal clotting when it asks what an **inhibitor stops**: that scores 1 mark at most.

BEYOND THE SPEC, BUT EXAMINED

- **DVT → pulmonary embolism**: a clot from a leg vein travels vena cava → right atrium → tricuspid valve → right ventricle → pulmonary artery, and lodges where the arteries narrow (WBI11 Jan 2026).
- **Thrombolytic drugs**: plasmin hydrolyses the peptide bonds in fibrin, dissolving a clot (WBI11 Jun 2019).
- Thromboplastin is stored in platelets, **separate from prothrombin**, so clotting can't start until it's needed, then it's released quickly (WBI11 Jun 2022).

SPEC 1.12

Risk factors for CVD

WHAT THIS SPEC POINT ASKS

Know how genetics, diet, age, gender, high blood pressure, smoking and inactivity increase the risk of cardiovascular disease.

A **risk factor** increases the **probability** of a disease; it doesn't make it certain. Risk factors **add up**: someone who smokes, is inactive and eats a high-saturated-fat diet carries all three risks at once.

Risk factor	How it raises the risk of CVD	Can you change it?
Genetics	inherited alleles can cause high blood cholesterol (e.g. faulty LDL receptors), high blood pressure, or weaker arteries	no
Age	arteries lose elasticity and plaque builds up over a lifetime	no
Gender	men are at higher risk at younger ages; **oestrogen** in women before menopause raises HDL (protective)	no
High blood pressure	damages the endothelium → starts atherosclerosis; the heart works harder	partly (diet, exercise, drugs)
Diet	high saturated fat → high **LDL**/blood cholesterol → plaque; high **salt** → water retained → raised blood pressure; excess energy → **obesity**	yes
Smoking	chemicals **damage the endothelium**; **nicotine** triggers adrenaline release → **arterioles constrict** → raised heart rate and **blood pressure**; nicotine makes **platelets sticky** → clots; **carbon monoxide** binds Hb, so less O_2 reaches heart and brain → heart works harder; fewer **antioxidants**; lower **HDL**	yes
Inactivity	linked to obesity, high blood pressure and high LDL; regular exercise raises HDL and lowers blood pressure	yes

EXAM TIP

Explain the **mechanism**, not just "it's bad for you": link each factor to one of endothelial damage, raised blood pressure, raised LDL or increased clotting.

MARK-SCHEME LANGUAGE

- Risk factor: "a variable that increases the chance of an event (the disease) happening".
- Non-lifestyle: age, sex, genetics. Lifestyle: smoking, BMI, salt, alcohol, inactivity. Always give the **direction**: "high BMI", "high salt intake". Bare "diet", "lifestyle" or "exercise" is ignored.
- Mechanisms that score: age → arteries "lose elasticity"; smoking → raises blood pressure and damages the endothelium; salt → raises blood pressure; high blood pressure → damages the endothelium.
- Describing a graph: "as the number of cigarettes smoked increases, the risk of CHD increases", "non-linear", there is still some risk with no smoking. Not "exponential".
- A study must adjust for confounders such as age and smoking, or it is "not valid". Reliability is ignored here.

Often asked as: classify factors as lifestyle or not, describe a risk graph, use a risk calculator, 6-mark comparing two risk calculators.

COMMON MISTAKE

Listing factors with no mechanism: that keeps a 6-mark answer at Level 1. "Poor diet" is too vague: say which part (high saturated fat, high salt) and how it acts.
Mixing up lifestyle and non-lifestyle factors. Two marks means two different points.

BEYOND THE SPEC, BUT EXAMINED

- **Risk calculators** (e.g. the Framingham score): add up the points for each factor and read off the % risk. Weaknesses: they leave out some factors, and people may guess their blood pressure or cholesterol (WBI11 Jun 2022).
- Women are partly protected **until the menopause** (WBI11 Jun 2023).

SPEC 1.13 Dietary antioxidants and CVD

WHAT THIS SPEC POINT ASKS

Understand the link between dietary antioxidants and the risk of CVD.

1.13 The idea

- Metabolism (and smoking) produces **free radicals**: unstable, highly reactive particles with an unpaired electron.
- Free radicals can **damage the endothelium** of arteries and **oxidise LDL**, making it more likely to be taken up into plaques. Both promote atherosclerosis.
- **Antioxidants** (e.g. **vitamins C and E**, polyphenols such as **flavonoids** in fruit, vegetables, tea and dark chocolate) react with and **neutralise free radicals**, so in theory they **reduce the risk** of CVD. Vitamin C is also needed to make collagen in artery walls.

1.13 The evidence is mixed

- **For**: many observational studies link high fruit and vegetable intake, or high blood vitamin C, with lower heart-attack rates.
- **Against**: large randomised trials and meta-analyses of **antioxidant supplements** found no clear benefit, and some suggest possible harm.
- People who eat lots of fruit and vegetables often also exercise more, smoke less and eat less saturated fat, so these **confounding variables** may explain the link. Correlation ≠ causation.

EXAM TIP

For "evaluate the evidence" questions: give the proposed mechanism, the supporting evidence, the conflicting evidence, then a balanced conclusion ("the evidence is inconclusive because…").

MARK-SCHEME LANGUAGE

Learn this chain word for word. The same mark scheme keeps coming back:

- "antioxidants reduce free radicals" (neutralise / stabilise / donate electrons to them)
- → less "damage to the endothelial lining" (less oxidative stress)
- → "reduced plaque / atheroma formation"
- → lower risk of CVD.
- Question about *less* antioxidant? Flip it: "fewer free radicals are reduced", so the damage isn't prevented.
- Why eat a food that isn't proven to work? It's "not harmful" and is a food source anyway. "It contains antioxidants" scores 0.

Often asked as: explain how antioxidants lower CVD risk, compare the antioxidant content of foods, design a study (6-mark).

COMMON MISTAKE

Stopping before plaque and CVD. "Reduces CVD" on its own just repeats the question. Don't drift into LDL: this chain is about free radicals and endothelial damage.

SPEC 1.14 Core Practical 2: vitamin C content of food and drink

WHAT THIS SPEC POINT ASKS

Investigate the vitamin C content of food and drink.

1.14 Principle

- **DCPIP** is a **blue** dye. **Vitamin C** (ascorbic acid) is a **reducing agent** that turns DCPIP **colourless** (reduces it).
- The more vitamin C in a solution, the **less** of it you need to decolourise a fixed volume of DCPIP.

CORE PRACTICAL 2: METHOD IN BRIEF

1. Put a fixed volume (e.g. **1 cm^3**) of DCPIP in a test tube.
2. Add a vitamin C solution of **known concentration** drop by drop from a burette or graduated pipette, **shaking gently** after each drop.
3. Stop when the blue colour **just disappears** (the end point); record the volume added.
4. Repeat (at least three times) and calculate a **mean**.

5. Repeat steps 1–4 with the fruit juice or food extract.
6. Optionally, test a range of known concentrations and plot a **calibration curve** of volume needed against concentration; read the unknown off the curve.

WORKED EXAMPLE

1 cm^3 of DCPIP is decolourised by **2.0 cm^3** of a **1.0 mg cm^{-3}** vitamin C standard, so it takes **2.0 mg** of vitamin C.
The same 1 cm^3 of DCPIP needs **8.0 cm^3** of orange juice.
So 8.0 cm^3 of juice contains 2.0 mg vitamin C → **0.25 mg cm^{-3}** of juice.
General rule: concentration of sample = (concentration of standard × volume of standard) ÷ volume of sample.

1.14 Variables and sources of error

- **Control**: volume and concentration of DCPIP, shaking, temperature, how the end point is judged (against a white tile).
- **Problems**: the end point is subjective; coloured juices mask the colour change (dilute them or use a colorimeter); vitamin C is **oxidised** by air and heat, so use fresh solutions and don't boil them.
- Safety: DCPIP is an irritant, so wear eye protection.

EXAM TIP

This practical hasn't been examined directly in any WBI11 paper from 2019 to 2026, but it's a named core practical, so still learn it. If it comes up, use the study-design language from 1.16: controlled variables make it **valid**; repeats and a mean make it **repeatable**.

SPEC 1.15 Interpreting data on illness and mortality

WHAT THIS SPEC POINT ASKS

Analyse and interpret quantitative data on illness and mortality to work out health risks, distinguishing correlation from causation and recognising conflicting evidence.

1.15 Key terms

- **Incidence**: number of **new** cases in a population over a set time (e.g. per 100 000 per year).
- **Prevalence**: number of people **with** the disease at a given time.
- **Mortality**: number of **deaths** from the disease (per 100 000 per year).
- **Relative risk**: risk in an exposed group ÷ risk in an unexposed group. A relative risk of 2.0 means twice as likely; 1.0 means no difference.

1.15 A three-step scaffold for data questions

1. **Describe**: state the trend, and **quote figures** from the data, with units (e.g. "relative risk rises from 1.0 to 2.4 between ages 40 and 70").
2. **Conclude**: say there is a **correlation** (a positive or negative association), only within the range of the data. Don't extrapolate to other ages, sexes or species.
3. **Evaluate**: is the link **causal**? Consider sample size, how the sample was chosen, controls, confounding variables, repeats, whether **error bars/standard deviations overlap** (non-overlapping bars suggest a **significant** difference), and whether a statistical test was done.

WORKED EXAMPLE: PERCENTAGE CHANGE

% change = (new − old) ÷ old × 100. Subtract first, then divide by the **old** value.
Deaths fall from 250 to 180 per 100 000: (180 − 250) ÷ 250 × 100 = **−28%**, a 28% decrease.
Give the d.p. or s.f. asked for, never a recurring decimal, and round numbers of people to whole numbers.

1.15 Correlation vs causation

- A **correlation** means two variables change together. It doesn't prove one causes the other: a **third (confounding) variable** might cause both.
- **Causation** is supported when there's a plausible **mechanism**, the relationship holds across many large, well-controlled studies, and changing the factor (e.g. in a trial) changes the outcome.
- **Conflicting evidence**: different studies may disagree because of different sample sizes or populations, uncontrolled variables, or bias. One study isn't enough; combine studies (meta-analysis) and look at the quality of each.

EXAM TIP

Always **use numbers from the data**, and use **correlation** or **association**, not "causes", unless the question gives you evidence of a mechanism or a controlled trial.

MARK-SCHEME LANGUAGE

- Correlation: "a change in one variable is reflected by a change in another variable" / "a relationship between two variables". "Causes", "affects", "results in" and "leads to" are rejected. Add "but this doesn't show that X causes Y" where it fits.
- Error bars / SD: not overlapping → "the difference is (probably) **significant**"; overlapping → "probably not significant"; none shown → "no error bars / SD, so significance can't be judged". Link error bars to **significance**, never to "reliability".
- Evidence *against* a link: people with the disease at low exposure, or high exposure without the disease.
- A hazard ratio or relative risk of 1.00 is the baseline: a "means of comparison". With hazard ratios, say "risk of death", not risk of the disease.
- Predicting beyond the data: use the word "**extrapolate**".
- Why figures are only estimates: undiagnosed people, unregistered births and deaths, people who are hard to reach or count.

Often asked as: define correlation, state two conclusions from a graph, give evidence for and against a link, comment on data with error bars, calculate a percentage change.

COMMON MISTAKE

Narrating bar by bar instead of giving the trend. Giving one conclusion and its reverse as "two conclusions". Describing only the rise when the data also fall. Misquoting values: read them carefully and give units.

BEYOND THE SPEC, BUT EXAMINED

- **Statistical tests**: a t-test (or Mann-Whitney U) checks whether two groups differ significantly; a chi-squared test compares observed and expected numbers (WBI11 Oct 2021).
- Pooled data give a useful overview but can hide differences between groups (WBI11 Jan 2025).

SPEC 1.16 Designing studies of health risk factors

WHAT THIS SPEC POINT ASKS

Evaluate the design of studies used to find health risk factors, including sample selection and sample size, so the data are valid and reliable.

1.16 What makes a good study

- **Large sample**: reduces the effect of chance and of anomalous individuals → results are more **repeatable**, and statistics can be done.
- **Representative, random sample**: matches the population you want to draw conclusions about (age, sex, ethnicity, lifestyle). Random selection avoids **selection bias**.
- **Control group / matched groups**: compare people with and without the risk factor who are otherwise similar (**matched** for age, sex, diet, etc.).
- **Control confounding variables**: other factors that affect the outcome (smoking, diet, exercise, age). Controlling them makes the study **valid**.
- **Exclude** people who already have the disease: the study is about its **development**.
- **Long enough** duration for the disease to develop.
- **Repeatable and reproducible**: the method is published so others can repeat it; results agree.
- **Avoid bias**: **double-blind** trials (neither participants nor researchers know who gets what), a **placebo** in drug trials, independent funding, peer review.
- Measurement: self-reported data (e.g. diet questionnaires) are unreliable, because people forget, lie or are embarrassed.

1.16 Types of study

Study	How it works	Strengths / weaknesses
Cohort	follow a large group over time; record exposure and who develops the disease	can show which came first; takes years and is expensive; people drop out
Case-control	compare people who already **have** the disease with similar people who **don't**, and look back at past exposure	quick and cheap; relies on memory (recall bias)
Randomised controlled trial	randomly assign people to treatment or placebo	best evidence for causation; not ethical for harmful factors like smoking

Use the mark schemes' words:

- **Valid** = the study measures what it claims to, because only the factor being studied differs. Controls and matched groups give validity.
- **Repeatable** = you get similar results when you repeat it; **reproducible** = others get similar results. A large sample and repeats give this.
- Avoid the bare words "reliable", "accurate" and "precise": say which of valid, repeatable or reproducible you mean.

EXAM TIP

"Evaluate the design" means **strengths and weaknesses**: comment on the sample (size, how chosen, who's included), controls, confounding variables and duration, and say *why* each matters.

MARK-SCHEME LANGUAGE

- Sample: "large groups" for "repeatability / reproducibility" and so statistics can be done. Participants "healthy / no known heart condition", "matched for age, sex, lifestyle". Side-effect trials use "healthy volunteers", not "patients".
- "control / placebo group"; "double blind … to avoid bias". In an animal injection study, the control gets **the liquid the drug was dissolved in**.
- Control the named variables "for validity"; if they aren't controlled, "the conclusions would not be valid".
- Duration: monitor the **incidence** of the disease over a long period (years for CVD, "because CVD develops slowly"); monitor side effects.
- Other criticisms: no control group, no statistics or error bars, no definition of "high" or "low" intake, "difficult to control all the variables".
- Say dose, volume or concentration, never "amount".

Often asked as: criticise a study (6-mark), design a study (6-mark), explain why a control or an exclusion was used, say whether a study is valid.

COMMON MISTAKE

Giving design points with no reason: each one needs a "so that…" (valid or repeatable). Inventing flaws the data don't show; "the groups must be equal sizes" isn't a valid criticism. Monitoring "risk" instead of **incidence**.
When asked whether a study is valid, **say which**, then give reasons.

BEYOND THE SPEC, BUT EXAMINED

- Add a point specific to the context, e.g. cigarettes per day, ex-smokers, self-diagnosed v clinically diagnosed. That's what lifts a 6-mark criticism to full marks (WBI11 Jun 2021).
- A small sample can be justified for a very rare disease (WBI11 Oct 2021).

SPEC 1.17

Perceived risk vs actual risk

WHAT THIS SPEC POINT ASKS

Understand why people's perception of risk often differs from the actual risk, including underestimating or overestimating risks from diet and other lifestyle factors in heart disease.

- **Actual risk** is the measured probability of harm, based on scientific data (e.g. "1 in 100 per year").
- **Perceived risk** is how risky people *feel* something is.

People overestimate risk when...	People underestimate risk when...
it's unfamiliar, dramatic or reported in alarming media stories	it's familiar, everyday and **enjoyable** (e.g. fast food, smoking)
harm is sudden and severe (e.g. plane crash)	harm is **delayed** by years (heart disease develops slowly)
they know someone affected	they have no personal experience of it
it's imposed on them	they feel in control ("I'll cut down later")

Applied to heart disease: many people underestimate the risk from a high-saturated-fat or high-salt diet, inactivity, obesity and smoking, because the effects are slow, invisible and linked to enjoyable habits. They may overestimate small risks reported dramatically in the media (e.g. a scare about one food) and ignore large, steady ones. People also misunderstand statistics, e.g. "doubles the risk" sounds huge even when the starting risk is tiny.

MARK-SCHEME LANGUAGE

- Underestimating: people "underestimate their mass / how much they smoke", "may not realise how much of a risk factor smoking can be". Self-reported data are unreliable.
- Risk calculators: they leave out other risk factors, and blood pressure or cholesterol values may be guessed.
- Consequences: overestimating → unnecessary lifestyle changes and stress; underestimating → no lifestyle change, testing or medication.
- A fair perception study uses people who don't know their own risk and aren't on CVD medication.

Often asked as: explain why someone misjudges their risk, compare perceived and actual risk on a bar chart, give the consequences of misjudging it.

COMMON MISTAKE

Describing a perceived-v-actual chart bar by bar. Give the overall pattern (e.g. perceived risk is generally lower than actual risk), then the exception.

SPEC 1.18 Cholesterol, HDL and LDL

WHAT THIS SPEC POINT ASKS

(i) Analyse data on how blood cholesterol, HDL and LDL levels might affect health. (ii) Know the evidence for a causal link between blood cholesterol (total and LDL) and CVD.

1.18 What they are

- **Cholesterol** is a lipid needed for **cell membranes** and to make steroid hormones and bile. It's insoluble, so it's carried in the blood as **lipoproteins** (lipid + protein).

	HDL (high-density lipoprotein)	LDL (low-density lipoprotein)
Made of	more protein; unsaturated fat + cholesterol	less protein; **saturated** fat + cholesterol
Carries cholesterol	**from tissues to the liver** (for removal or recycling)	**from the liver to body cells** (binds LDL receptors)
Effect	**lowers** blood cholesterol; may remove it from plaques	when receptors are saturated, LDL stays in the blood and is deposited in artery walls → **atheroma**
Nickname	"good"	"bad"

- Diets high in **saturated fat** raise LDL; **unsaturated** fats raise HDL and lower LDL.
- The **ratio** of LDL to HDL (or total cholesterol to HDL) predicts risk better than either one alone: high LDL with low HDL is the worst combination.

1.18 Evidence that high cholesterol *causes* CVD

- A clear **mechanism**: LDL cholesterol accumulates in damaged artery walls and forms plaques (1.10).
- **Large studies** in many populations show a strong positive correlation between total/LDL cholesterol and CVD.
- **Intervention trials**: lowering LDL with **statins** (in randomised, placebo-controlled trials) reduces heart attacks and deaths; the bigger the fall in LDL, the bigger the fall in risk.
- **Genetic evidence**: people with **familial hypercholesterolaemia** (an inherited fault in LDL receptors) have very high LDL from birth and get CVD young.

1.18 Reading the data

- Quote values; compare groups; look at **error bars**: if the standard deviation bars **overlap**, the difference may **not be significant**.
- Consider confounders: people with high LDL may also smoke, be overweight or inactive.

MARK-SCHEME LANGUAGE

- "the ratio / balance of LDL : HDL determines the risk". Risk is highest with high LDL **and** low HDL.
- "HDL transports cholesterol to the liver"; "LDL forms the plaque".
- LDL structure: the protein and phospholipid heads on the outside are **hydrophilic**, so they interact with the plasma; the cholesterol is surrounded inside.
- Data (e.g. a 3-D bar chart): at every HDL level, risk rises as LDL rises; at every LDL level, risk falls as HDL rises. Quote the evidence, don't just restate the conclusion.

Often asked as: explain HDL/LDL data, 6-mark on why one measurement alone doesn't predict risk, LDL structure and transport.

COMMON MISTAKE

Writing that HDL "is good cholesterol". Cholesterol is the same molecule: HDL and LDL differ in the **lipoprotein** carrying it and **which direction** they take it.
Leaving out how LDL mixes with the plasma. Calling the LDL receptor a channel or an enzyme. Describing the data without explaining it scores 1.

BEYOND THE SPEC, BUT EXAMINED

- **LDL receptors**: LDL binds a receptor with a complementary shape and is taken into the cell by **endocytosis**. In familial hypercholesterolaemia a mutation changes the receptor's tertiary structure, so LDL can't bind and stays in the blood (WBI11 Oct 2023).
- Statins work by blocking **HMG-CoA reductase**, the liver enzyme that makes cholesterol (WBI11 Oct 2024).
- **Fibre** reduces the absorption of cholesterol from the gut; plant sterols also lower it (WBI11 Jan 2026).

SPEC 1.19 Using diet, exercise and smoking knowledge to reduce CHD risk

WHAT THIS SPEC POINT ASKS

Understand how people use scientific knowledge about diet (including obesity indicators such as BMI and waist-to-hip ratio), exercise and smoking to reduce their risk of coronary heart disease.

1.19 Energy balance

- **Energy in** (food) > **energy out** (metabolism + activity) → the surplus is stored as fat → weight gain and **obesity**.
- Energy in < energy out → body stores used → weight loss.
- Obesity raises CHD risk: raised blood pressure, raised LDL, and a higher risk of type 2 diabetes.

1.19 Obesity indicators

- **BMI = mass (kg) ÷ height^2 (m^2)**.

BMI	Category
below 18.5	underweight
18.5–24.9	healthy
25–29.9	overweight
30 and over	obese

WORKED EXAMPLE: BMI AND WAIST-TO-HIP RATIO

Mass 84 kg, height 1.75 m: BMI = 84 ÷ 1.75^2 = 84 ÷ 3.0625 = **27.4** → overweight.
Waist 96 cm, hips 102 cm: waist-to-hip ratio = 96 ÷ 102 = **0.94** (or 0.94 : 1). That's above about 0.90 (men) or 0.85 (women), so the risk is raised.
Rearranged: mass = BMI × height^2. A BMI of 30 at 1.75 m is 30 × 3.0625 = **91.9 kg**.
Height in cm? Convert to m **before** squaring.

- **Limitations of BMI**: doesn't distinguish fat from muscle (athletes can score "overweight"), and ignores **where** fat is stored.
- **Waist-to-hip ratio** measures **abdominal (central) fat**, which is more closely linked to CHD, so it can be a better predictor.
- Other obesity indicators: **waist circumference**, **waist-to-height ratio**, **skinfold thickness**, **% body fat**. Weight on its own isn't an indicator of obesity.

1.19 How people act on the science

- **Diet**: less saturated fat and salt, more fruit, vegetables and fibre; reading food labels (traffic-light labels for fat, saturated fat, sugar and salt); controlling energy intake.
- **Exercise**: regular aerobic activity lowers blood pressure and LDL, raises HDL and helps control weight.
- **Smoking**: stopping reduces risk quickly (support services, warnings on packets, advertising bans, smoke-free laws).

MARK-SCHEME LANGUAGE

- Indicators other than BMI: "waist-to-hip ratio", "waist (circumference)", waist-to-height ratio, "skinfold thickness", % body fat. Weight and "pinch test" are ignored, and so is BMI when the question says "other than BMI".
- BMI = mass (kg) ÷ height^2 (m^2). 25 to under 30 = overweight; 30 or more = obese. WHR = waist ÷ hip.
- Diet changes, each with its reason: less saturated fat / cholesterol → less atheroma; less salt → lower blood pressure, less endothelial damage; more antioxidants → reduce free radicals; more fibre → less cholesterol absorbed; fewer calories.
- Weight loss: "use more energy than is consumed", so eat less **and** exercise more. It must be a change: "regular exercise" alone is ignored.

Often asked as: calculate and classify a BMI, rearrange the BMI or WHR formula, name an indicator other than BMI, recommend diet changes with reasons.

COMMON MISTAKE

Giving a risk factor when an **indicator** was asked for, or giving only one. "Change your diet" is too vague: name the change and its direction.
Putting smoking in a "diet" answer. Giving the reason as just "increases the risk of CVD": say how (e.g. raises blood pressure).

BEYOND THE SPEC, BUT EXAMINED

- **Skinfold callipers** measure fat under the skin. Limits: fat isn't spread evenly, exercise can replace fat with muscle, and readings mislead in pregnancy (WBI11 Jun 2021).
- WHR measuring errors: the tape pulled too tight at the waist, or the person breathing in (WBI11 Jun 2021).

SPEC 1.20 Treatments for CVD: benefits and risks

WHAT THIS SPEC POINT ASKS

Know the benefits and risks of treatments for CVD: antihypertensives, statins, anticoagulants and platelet inhibitors.

For each drug, learn the chain: **what it does → what that prevents in 1.10/1.11 → its risks**.

Treatment	How it works	Benefit	Risks / side effects
Antihypertensives (e.g. ACE inhibitors, beta-blockers, diuretics, calcium-channel blockers)	lower blood pressure: widen blood vessels, slow the heart, or reduce blood volume	less endothelial damage → less atherosclerosis; lower risk of stroke and heart attack	dizziness, fainting, tiredness, palpitations, kidney problems; may need several drugs
Statins	block a liver enzyme used to make cholesterol → lower blood **LDL**	fewer plaques form → lower risk of heart attack and stroke	muscle aches and weakness, liver damage, digestive upsets; small increased risk of diabetes; lifelong use
Anticoagulants (e.g. warfarin, heparin)	interfere with clotting factors (e.g. less prothrombin made)	fewer clots → lower risk of thrombosis, stroke, DVT	**excessive bleeding**/haemorrhage when injured; dose must be monitored; interacts with other drugs and some foods
Platelet inhibitors (e.g. aspirin, clopidogrel)	stop platelets sticking together	fewer clots form at damaged arteries	**bleeding**, including stomach bleeding and ulcers; allergic reactions

Anticoagulants vs platelet inhibitors: both reduce clotting, but anticoagulants act on **clotting factors** in the plasma, while platelet inhibitors act on **platelets**.

General issues with all drug treatment: side effects, the cost of lifelong medication, and the fact that drugs treat the risk without changing the lifestyle causing it.

EXAM TIP

"Benefits and risks" questions are marked as pairs. For each drug give **one benefit linked to the disease process** and **one specific risk**. "Bleeding" alone is too vague: say *excessive bleeding if injured* or *internal/stomach bleeding*.

MARK-SCHEME LANGUAGE

- Antihypertensives lower blood pressure; statins lower blood cholesterol (LDL); anticoagulants prevent clotting; platelet inhibitors reduce clotting.
- Anticoagulant risks: "bleeds too easily / internally", bruising, nosebleeds, heavy periods. Dose too high → "clotting time too long … too much blood could be lost"; too low → clots form → heart attack, stroke, embolism or DVT.
- Statin side effects: muscle pain, headache, nausea, liver damage, fatigue, diabetes. Learn 2–3 for each drug class.
- Why use several drugs: "many factors cause CVD", so "different drugs treat different conditions".

Often asked as: match a drug to its use (MCQ), name a risk or side effect, explain why a warfarin dose is monitored, calculate a percentage change in clotting time.

COMMON MISTAKE

Confusing a **side effect** (e.g. headache, bruising) with a **risk** (e.g. a serious bleed, or a clot if the dose is too low). Not knowing which drug treats which condition.
Calling warfarin a non-competitive inhibitor. Writing "gene" when you mean **allele** caps a pharmacogenomics answer at Level 1.

BEYOND THE SPEC, BUT EXAMINED

- **Warfarin** has a similar shape to vitamin K and **competitively** inhibits the enzyme that recycles it, so less prothrombin is made. Extra vitamin K competes with warfarin for the active site (WBI11 Oct 2020).
- **INR** (international normalised ratio) is used to monitor warfarin: too high → bleeding risk, too low → clotting risk, so it is kept within a safe range (WBI11 Oct 2022).
- **Pharmacogenomics**: people with alleles for slow warfarin breakdown need a lower dose (WBI11 Oct 2024).

Topic 2 – Membranes, Proteins, DNA and Gene Expression

SPEC 2.1 Gas exchange surfaces, Fick's law and the lung

WHAT THIS SPEC POINT ASKS

(i) Know the properties of gas exchange surfaces: large surface area to volume ratio, thin surface, concentration difference. (ii) Understand how these set the rate of diffusion, and calculate it with Fick's law. (iii) Understand how the mammalian lung is adapted for rapid gas exchange.

2.1 The three properties

1. **Large surface area** (relative to volume): more area for molecules to cross at once.
2. **Thin surface / short diffusion distance**: molecules have less far to travel.
3. **Steep concentration gradient**: kept steep by bringing in fresh supply on one side and carrying away on the other.

Big organisms have a **larger surface area** but a **smaller SA:V ratio**, so they need specialised exchange surfaces (lungs, gills, leaves) and a transport system (1.6).

2.1 Fick's law

$$\text{rate of diffusion} \propto (\text{surface area} \times \text{difference in concentration}) \div \text{thickness of surface}$$

- Double the area → double the rate. Double the concentration difference → double the rate. **Double the thickness → halve the rate.**

WORKED EXAMPLE: FICK'S LAW

Alveolus A: area 70 units, concentration difference 10 units, thickness 0.5 μm → rate ∝ (70 × 10) ÷ 0.5 = **1400**.
Alveolus B after thickening (e.g. fibrosis): thickness 1.0 μm → (70 × 10) ÷ 1.0 = **700**.
Doubling the thickness halved the rate. Questions often ask for the **percentage change**: (700 − 1400) ÷ 1400 × 100 = **−50%**.

2.1 How the lung is adapted

Feature	How it speeds up gas exchange
Millions of **alveoli** (and branching bronchioles)	**huge surface area** (about 70 m^2 in humans)
Alveolar wall is **one cell thick** of **flattened (squamous) epithelium**	**short diffusion distance**
Capillary wall is one cell thick, pressed against the alveolus; red cells squeezed against the wall	short distance (total barrier about 0.5 μm)
Dense **capillary network** around each alveolus; blood constantly flowing	removes O_2 and brings CO_2 → **maintains the concentration gradient**
Ventilation (breathing)	replaces alveolar air → keeps O_2 high and CO_2 low in the alveoli → maintains the gradient
Moist lining	gases dissolve before diffusing across
Elastic fibres	stretch on breathing in, recoil to push air out
Surfactant	stops alveoli collapsing (beyond spec)

Airways: the **trachea** and **bronchi** are held open by **cartilage**; ciliated epithelium and **goblet cells** (mucus) trap and remove particles; **bronchioles** have smooth muscle to change their diameter.

MARK-SCHEME LANGUAGE

Fick's law: "rate ∝ (surface area × difference in concentration) ÷ thickness of surface".
Link every feature to rate: "alveoli → **large surface area** → faster diffusion" · "one cell thick / squamous epithelium → **short diffusion distance**" · gradient "**maintained by ventilation / blood flow**". A list of features with no links scores 1.
Disease (emphysema, CF): "smaller surface area / longer distance / lower gradient" → "diffusion of oxygen is **slower**". **Ignored**: "less gas exchange", "less efficient". **Not accepted**: "no diffusion".
Factors other than distance: surface area, concentration gradient, permeability, temperature. "SA:V" is **not accepted** here.
Often asked as: name the law; predict the effect of halving or doubling one term; explain lung adaptations; explain how a disease or an unfamiliar surface (egg shell, amphibian skin) changes the rate.

BEYOND THE SPEC, BUT EXAMINED

- **Temperature** and the **diffusion coefficient** (gases diffuse faster through air than through a membrane) also affect the rate (WBI11 Jan 2021).
- **Behaviour can keep the gradient steep**: a salamander rocks its body to stir the water over its skin, replacing deoxygenated water (WBI11 Oct 2022).
- **Unfamiliar surfaces** (amphibian skin, egg-shell pores): apply the three Fick's-law terms to whatever the question shows you (WBI11 Jan 2021).

COMMON MISTAKE

Saying "the alveoli have thin **cell walls**". Animal cells don't have cell walls: say "the **alveolar wall** is one cell thick". Also, don't confuse "large surface area" with "large SA:V ratio".
Calling it Bohr's or Boyle's law. It's **Fick's law**.
Writing "more diffusion" or "more efficient". Say "**faster** diffusion" or "a higher **rate** of diffusion".

EXAM TIP

For "explain how the lung is adapted" answers, give the **feature + which Fick's-law term it affects**: "many alveoli → large surface area → faster diffusion".

SPEC 2.2

Cell membranes and the fluid mosaic model

WHAT THIS SPEC POINT ASKS

(i) Know the structure and properties of cell membranes. (ii) Understand how models such as the fluid mosaic model are interpretations of data, used to develop scientific explanations of membrane structure and properties.

2.2 The parts of the membrane

- **Phospholipid bilayer**: each phospholipid has a **hydrophilic phosphate head** (faces the water inside and outside the cell) and **two hydrophobic fatty acid tails** (point inwards, away from water). The bilayer forms spontaneously in water.
- **Proteins**:
  - **Intrinsic** (integral) proteins span the bilayer: **channel proteins** and **carrier proteins** (2.5).
  - **Extrinsic** (peripheral) proteins sit on one surface: enzymes, receptors, structural supports.
  - A protein's position depends on where its hydrophobic and hydrophilic R groups are.
- **Cholesterol**: sits between phospholipids, with its small polar **OH group next to the phosphate heads** and the rest of the molecule **among the fatty acid tails**. It **regulates fluidity**: it makes the membrane less fluid at high temperatures and stops it becoming too rigid at low temperatures; adds stability.
- **Glycoproteins and glycolipids**: carbohydrate chains on the outer surface, used for **cell recognition**, as receptors (e.g. for hormones) and in cell adhesion.

2.2 Why "fluid mosaic"?

- **Fluid**: phospholipids and many proteins move **sideways** within the layer.
- **Mosaic**: proteins are scattered through the bilayer like tiles in a mosaic.

2.2 Properties

- **Partially permeable**: small, non-polar molecules (O_2, CO_2) and water pass straight through the bilayer. Ions and large or polar molecules (glucose, amino acids) need **transport proteins**, because the hydrophobic core repels them.

2.2 How the model was built from data (models are interpretations)

Model	Evidence it explained	Why it was replaced
Lipid bilayer (Gorter and Grendel, 1925)	lipid extracted from red blood cell membranes covered **twice** the cells' surface area → two layers	didn't explain how polar substances cross, or the role of proteins
Protein–lipid sandwich (Davson–Danielli, 1930s)	electron micrographs showed two **dark lines** (taken to be protein) with a **light band** (lipid) between	membrane proteins turned out to be varied and often hydrophobic; a protein coat would block lipid-soluble molecules
Fluid mosaic (Singer–Nicolson, 1972)	**freeze-fracture** micrographs showed proteins **embedded in and spanning** the bilayer; labelled proteins from two fused cells **mixed** quickly, so they move	still accepted, refined by later data

The point: scientists propose a model that fits the data; **new evidence** (often from new technology such as electron microscopy) can lead to the model being **modified or replaced**. A model is a representation, not the real thing.

EXAM TIP

Say what **evidence** supported or overturned each model: that's where most marks are. The names have also been examined (a whole question on the history in Jan 2025), so learn them in order (see the box below).

MARK-SCHEME LANGUAGE

Bilayer: "hydrophilic / polar **heads** interact with water" on **both sides**; "hydrophobic / non-polar **tails** are repelled by water", so they point inwards. Describing the arrangement without the reason scores 1.
Drawing a phospholipid: **glycerol + two fatty acids + phosphate**, with every bond line touching.
Cholesterol position: "the polar (OH) region interacts with the **phosphate heads** and the non-polar region is **amongst the lipid tails**".
Fluidity: a higher temperature "increases the **movement of phospholipids**" because "interactions between fatty acid chains break"; cholesterol "**fills the gaps** between the phospholipid tails", so it "decreases fluidity". **Not accepted**: "flexible", "strong".
"Fluid" = phospholipids move; "mosaic" = proteins scattered through the bilayer. "The bilayer moves" isn't credited.
Often asked as: draw or label a phospholipid; explain the bilayer arrangement; read fluidity or permeability graphs with and without cholesterol; compare Davson–Danielli with the fluid mosaic model.

BEYOND THE SPEC, BUT EXAMINED

- **History of the model, with names** (WBI11 Jan 2025): Overton (lipid-soluble substances enter cells faster, so membranes contain lipid) → Langmuir (lipid monolayer) → Gorter and Grendel (bilayer) → Davson–Danielli (protein–lipid sandwich) → Robertson (unit membrane) → Singer and Nicolson (fluid mosaic, 1972).
- **Cholesterol as a fluidity buffer** (WBI11 Jun 2025): with cholesterol, fluidity rises steadily (roughly linearly) with temperature; without it, the curve is S-shaped with a sharp rise. Cholesterol **raises** fluidity at low temperatures and **lowers** it at high temperatures.
- **Cold-adapted membranes** (fish in cold water): more **unsaturated** and **shorter** fatty acid chains pack less tightly, so the membrane stays fluid. Fluidity is needed for endocytosis, exocytosis, cell shape changes and cell fusion (WBI11 Jun 2022).
- **Checking a micrograph against the model** (WBI11 Jan 2021): the bilayer is about **5 nm** thick. Actual size = image size ÷ magnification, in the same units: a membrane 0.5 mm thick on a ×100 000 micrograph is 0.5 ÷ 100 000 = 5 × 10^{-6} mm = **5 nm**.

COMMON MISTAKE

Drawing **three** fatty acids on a phospholipid (that's a triglyceride), or joining the fatty acids to the phosphate.
Saying the heads face water without saying **both sides** of the membrane.
"The phospholipids melt" or "the bilayer moves". Say the **phospholipids move more**.
Describing cholesterol's **function** when the question asks for its **position**.
For cold-water fish, stopping at "keeps the membrane fluid" without the fatty-acid reason, or saying it "keeps the fish warm".

SPEC 2.3

Core Practical 3: membrane permeability

WHAT THIS SPEC POINT ASKS

Investigate membrane properties, including how alcohol concentration and temperature affect membrane permeability.

2.3 Principle

Beetroot cells contain a red pigment (betalain) in their vacuoles. When membranes (tonoplast and cell surface membrane) are damaged or more permeable, pigment **leaks out** into the surrounding water. More pigment → darker red → higher **absorbance** in a colorimeter.

CORE PRACTICAL 3: METHOD IN BRIEF

1. Cut beetroot cores with a **cork borer**, then slice into pieces of **equal size** (same length and diameter, so the same surface area) with a scalpel on a tile.
2. **Rinse** the pieces in distilled water to remove pigment released from cells cut open, then blot dry.
3. **Temperature**: put each piece in the same volume (e.g. 5 cm^3) of distilled water in tubes held in water baths at a range of temperatures (e.g. 0, 20, 40, 60, 80 °C) for the same time (e.g. 30 minutes). Let the water reach temperature first.
4. **Alcohol**: put pieces in the same volume of different ethanol concentrations (e.g. 0, 10, 20, 30, 40, 50%) at the **same temperature** (room temperature) for the same time.
5. Remove the beetroot, shake the tube, and transfer some of the liquid to a cuvette.

6. Zero the colorimeter with distilled water (or the alcohol solution) using a **blue-green filter**, then read the **absorbance** of each sample.
7. Repeat at least three times at each value and calculate means.

2.3 Explaining the results

- **Temperature rises** → phospholipids gain kinetic energy and move more, so the bilayer becomes more fluid and gaps appear; above about 40–50 °C **membrane proteins denature**, leaving holes → much more pigment leaks out. (Freezing can also raise permeability: ice crystals pierce membranes.)
- **Alcohol concentration rises** → ethanol is a non-polar solvent that **dissolves the lipids** in membranes and disrupts the bilayer → more pigment leaks.

2.3 Limitations and fixes

- Pieces not identical → measure carefully; use the same cork borer.
- Pigment varies across the root → take all pieces from the same root and average.
- Colorimeter cuvettes may differ or be scratched → use the same cuvette, rinsed.
- Time in water baths varies → start the timer as each piece goes in.

MARK-SCHEME LANGUAGE

"same surface area/size of beetroot pieces", "rinse to remove pigment from cut cells", "same volume of water", "same time", "colorimeter / absorbance", "proteins denature", "phospholipids move more / bilayer more fluid", "alcohol dissolves (phospho)lipids"
Temperature mechanism: more kinetic energy → "interactions between fatty acid chains break" → phospholipids move more → more permeable.
Data with two variables (temperature and cholesterol): permeability rises with temperature and falls as cholesterol rises, and "cholesterol has a **greater effect at higher temperatures**". That last point is the one most often missed.
Often asked as: the beetroot method itself wasn't set in WBI11 papers from 2019 to 2026, but permeability and fluidity data questions were, so learn the explanations as well as the method.

BEYOND THE SPEC, BUT EXAMINED

Liposome volume (WBI11 Jan 2019): $V = 4/3\pi r^3$ using the **radius**. A liposome 100 nm across has r = 50 nm, so $V = 4/3 \times \pi \times 50^3 =$ **5.2×10^5 nm^3**.

COMMON MISTAKE

Using the **diameter** instead of the radius in sphere formulas: the commonest maths slip.
With two variables, describe each effect **and** how one changes the effect of the other (the interaction).

SPEC 2.4 Osmosis and water potential

WHAT THIS SPEC POINT ASKS

Understand osmosis as the net movement of free water molecules through a partially permeable membrane, down a water potential gradient.

2.4 Definitions

- **Osmosis** is the **net movement of free water molecules** from a region of **higher water potential** to a region of **lower water potential**, through a **partially permeable membrane**.
- **Free water molecules**: when a solute dissolves, water molecules cluster around it and are **no longer free** to move across the membrane. More solute → fewer free water molecules → lower water potential.
- **Water potential (ψ)** is measured in **kPa**. **Pure water has the highest possible water potential: 0 kPa.** Any solution has a **negative** value; the more concentrated, the more negative.

WHICH WAY DOES WATER MOVE?

Cell A: ψ = −400 kPa. Cell B: ψ = −700 kPa. Water moves **from A to B** (from −400 to the more negative −700), i.e. down the water potential gradient.

2.4 What happens to cells

Surrounding solution compared with the cell	Animal cell (e.g. red blood cell)	Plant cell
Higher ψ (more dilute; e.g. pure water)	water enters → swells → may **burst (lysis)**	water enters → vacuole swells → cell becomes **turgid**; the cell wall stops it bursting
Same ψ	no net movement	no net movement; cell **flaccid** if already low on water
Lower ψ (more concentrated)	water leaves → cell **shrinks** (crenation)	water leaves → vacuole shrinks → membrane pulls away from the wall: **plasmolysis**

RECOMMENDED PRACTICAL: TISSUE WATER POTENTIAL

1. Cut equal-sized potato (or other plant tissue) cylinders, blot and **weigh** each.
2. Place each in a different sucrose concentration (e.g. 0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm^{-3}) for the same time (e.g. 24 h) at the same temperature.
3. Remove, **blot dry** (to remove surface solution) and reweigh.
4. Calculate **% change in mass** for each.
5. Plot % change in mass against concentration. Where the line **crosses zero** (no net gain or loss) the solution has the **same water potential as the tissue**.

WORKED EXAMPLE: % CHANGE IN MASS

Start mass 2.50 g, end mass 2.20 g. % change = (2.20 − 2.50) ÷ 2.50 × 100 = **−12%** (a loss: the solution had a lower ψ than the tissue).

Use **% change**, not mass change, because the starting masses differ.

MARK-SCHEME LANGUAGE

Always **compare**: "pond water has a **higher water potential than** the cytoplasm" → "so water moves in **by osmosis**". Hypotonic / hypertonic are accepted only as comparisons.
Animal cell in water: "**no cell wall**" → "can't stop water entering" → "swells / pressure increases" → "**bursts**".
Explaining the definition part by part: "free" = some water molecules are held by solute molecules; "partially permeable" = lets some molecules through but not others; "down a water potential gradient" = from higher to lower ψ.
The **total** concentration of all the solutes (all the ions) sets the water potential.
Ignored: "water concentration", "isotonic". **Wrong**: "the solution moves in".
Often asked as: explain water movement for a named organism or tissue; explain each part of the osmosis definition; 6-mark data questions (contractile vacuoles, crabs moving into sea water, osmotic dehydration of fruit).

BEYOND THE SPEC, BUT EXAMINED

- **Contractile vacuole** (Paramecium in pond water): water keeps entering by osmosis; the vacuole collects it and **pumps it out using ATP**. It doesn't stop water getting in (WBI11 Jan 2020).
- **Osmoconformers** keep their internal water potential the same as sea water, so there's no net osmosis, no shrinking or bursting, and no ATP spent (WBI11 Jun 2023).
- Animals moving into sea water can lower the water potential of their cells with **amino acids** from hydrolysed muscle protein (WBI11 Jan 2022).

COMMON MISTAKE

Writing "water moves from a high to a low **concentration**" or "down a concentration gradient". Use **water potential** every time. Don't say the solute moves: it's the water that moves.
Saying "hypotonic" without saying **compared with what**. Not naming **osmosis** when you explain the water movement.
Asked to explain the definition? Explain each part; don't just write the definition out again.

SPEC 2.5

Passive and active transport, endocytosis and exocytosis

WHAT THIS SPEC POINT ASKS

(i) Understand passive transport (diffusion, facilitated diffusion), active transport (including ATP as the immediate source of energy), endocytosis and exocytosis. (ii) Understand the roles of carrier and channel proteins in membrane transport.

2.5 Summary table

	Diffusion	Facilitated diffusion	Active transport	Endocytosis / exocytosis
Direction	down a concentration gradient	down a concentration gradient	**against** a concentration gradient	bulk movement (either direction)
Energy from ATP?	no (passive)	no (passive)	**yes**	**yes**
Protein needed?	no	**channel** or **carrier**	**carrier** only	no (membrane vesicles)
What moves	small, non-polar (O_2, CO_2); water	ions, polar molecules (glucose, amino acids)	ions, glucose, amino acids	large molecules, particles, whole cells
Example	O_2 into alveolar blood	glucose into red blood cells; Cl^- through CFTR channels	mineral ions into root hairs; glucose reabsorbed in kidney	phagocytosis of bacteria; secretion of enzymes or hormones

2.5 The proteins

- **Channel proteins**: water-filled pores lined with hydrophilic R groups. **Ions** and small polar molecules diffuse through. Each is specific; many can open and close ("gated").
- **Carrier proteins**: the molecule binds to a specific **binding site**, the protein **changes shape**, and the molecule is released on the other side. In **facilitated diffusion** this happens down the gradient. In **active transport** the shape change is driven by **ATP hydrolysis**.

2.5 Active transport and ATP

1. The molecule or ion binds to the carrier protein on one side.
2. **ATP is hydrolysed** (ATP → ADP + P_i), releasing energy (and the phosphate attaches to the carrier).
3. The carrier changes shape and releases the molecule on the other side, **against** its concentration gradient.
4. The phosphate is released and the carrier returns to its original shape.

ATP is the *immediate* source of energy because its hydrolysis releases energy in **small, manageable amounts**, in **one step**, exactly where it's needed. ATP is made by respiration, so anything that stops respiration (cyanide, no O_2, low temperature) stops active transport.

2.5 Bulk transport

- **Endocytosis**: the cell surface membrane **folds inwards** around a substance and pinches off as a **vesicle** inside the cell (e.g. phagocytes engulfing bacteria). Needs ATP.
- **Exocytosis**: **vesicles** (often from the Golgi apparatus) move to the cell surface, **fuse** with the membrane and release their contents outside (e.g. digestive enzymes, insulin, mucus). Needs ATP.

EXAM TIP

Down a gradient with a protein = **facilitated diffusion**. Active transport uses **carrier proteins**, never channels. To show a process is active, look for: movement **against** a gradient, or the rate falling when **respiration is inhibited**.

MARK-SCHEME LANGUAGE

Active transport: "**against** the concentration gradient", "needs **ATP** and a **carrier** protein". State the direction.
Movement through a channel is "**facilitated diffusion**", not "diffusion".
Only **water** moves by osmosis; osmosis moves water, not solute.
Which molecule crosses how: water is "**small** enough" to cross the bilayer (or uses aquaporins); non-polar molecules (O_2, steroids) "diffuse **through the phospholipid tails**"; glucose is **large and polar**, so it can't pass the hydrophobic tails and uses **carrier proteins**; ions use channels or active transport.
Endocytosis: "the membrane **surrounds** the substance". "Pinocytosis" / "phagocytosis" are ignored as names.
Often asked as: Venn diagrams or tick tables of the processes; cloze passages; explain an uptake curve; 6-mark "how do these four molecules cross the membrane".

BEYOND THE SPEC, BUT EXAMINED

- Channel proteins have **polar R groups lining the pore** and **non-polar R groups facing the fatty acid tails** (WBI11 Jan 2024).
- Uptake graphs (WBI11 Oct 2019): diffusion **levels off at equilibrium**; active transport carries on **past** equilibrium, so the inside ends up more concentrated. At a lower temperature, diffusion is slower (less steep) but levels off at the same value.

COMMON MISTAKE

Naming the process without the reason, e.g. "active transport" without "against the gradient, using ATP".
Leaving out that water is **small**, or that non-polar molecules pass **through the tails**.
Sketching a diffusion curve that never levels off.

SPEC 2.6

Amino acids and protein structure

WHAT THIS SPEC POINT ASKS

(i) Know the basic structure of an amino acid. (ii) Understand how polypeptides form by condensation, making peptide bonds. (iii) Understand how primary structure determines secondary, tertiary (3-D) structure and properties, including globular vs fibrous proteins and the bonds involved. (iv) Know the structure of a globular protein (haemoglobin) and a fibrous protein (collagen) and how it relates to function.

2.6 Amino acids and peptide bonds

- Every amino acid has a central carbon attached to: an **amine group (–NH_2)**, a **carboxyl group (–COOH)**, a **hydrogen**, and an **R group** (side chain). There are **20** amino acids, differing only in the R group. (Specific R groups don't need learning.)
- **Condensation**: the –OH of one amino acid's carboxyl group and an –H from the next one's amine group are removed as **water**, forming a **peptide bond** (–CO–NH–). Two make a **dipeptide**; many make a **polypeptide**.
- **Hydrolysis** (by protease enzymes) adds water and breaks the peptide bond.
- **Calculation**: n amino acids are joined by n − 1 peptide bonds, so the mass of a polypeptide = total mass of the amino acids − 18 × (n − 1), one water lost per bond.

2.6 Four levels of structure

Level	What it is	Bonds holding it
Primary	the **sequence** of amino acids (set by the gene)	peptide bonds
Secondary	the chain folds into an **α-helix** or a **β-pleated sheet**	**hydrogen bonds** between C=O and N–H groups of the backbone
Tertiary	the whole chain folds into a specific **3-D shape**	bonds between **R groups**: hydrogen bonds, **ionic bonds**, **disulfide bridges** (between cysteines; strong, covalent), hydrophobic interactions
Quaternary	two or more polypeptide chains together (sometimes with a prosthetic group, e.g. haem)	same as tertiary

Why primary structure matters: the sequence decides which R groups are where, so it decides where the bonds form, so it decides the **3-D shape**, which decides the **function** (e.g. the active site of an enzyme). Change one amino acid (a mutation, 2.14) and the shape can change; e.g. in **sickle cell anaemia** one polar amino acid in haemoglobin (**glutamic acid**) is replaced by a non-polar one (**valine**), and the haemoglobin clumps.

2.6 Globular vs fibrous

	Globular (e.g. haemoglobin, enzymes, insulin)	Fibrous (e.g. collagen, keratin)
Shape	compact, spherical	long, thin strands
R groups	**hydrophobic inside, hydrophilic outside**	repetitive sequence
Solubility	**soluble** in water	**insoluble**
Role	metabolic: transport, enzymes, hormones	**structural**: strength

2.6 Haemoglobin (globular)

- **Quaternary** structure: **four** polypeptide chains (2 α, 2 β), each with a **haem** prosthetic group containing **Fe^{2+}** that binds O_2.
- Hydrophilic R groups on the outside → **soluble** → carried in solution inside red blood cells.
- Its shape changes slightly as O_2 binds (cooperative binding, 1.9).

2.6 Collagen (fibrous)

- Three polypeptide chains wound into a **triple helix** (a rope-like structure).
- Every **third amino acid is glycine** (the smallest), so the chains can pack **tightly**. The sequence repeats and is rich in **proline** and **hydroxyproline**.
- **Hydrogen bonds** between the chains, and **covalent cross-links** between neighbouring molecules, form **fibrils**; the ends of molecules are **staggered** so there are no weak points. Fibrils bundle into **fibres**.
- Result: very high **tensile strength** and insoluble → found in tendons, skin, bone and **artery walls**.

MARK-SCHEME LANGUAGE

Haemoglobin: "**four haem groups**" (iron only credited if linked to haem) · "globular, **polar/hydrophilic R groups outside**" · "**soluble in the cytoplasm** of red blood cells" (**"soluble in blood" is not accepted**) · "cooperative binding between subunits"
"sequence of amino acids" (primary) · "hydrogen bonds **between amino acids**" (secondary; **not** "between R groups") · "bonds **between R groups**: hydrogen, ionic, disulfide" (tertiary; naming a wrong bond loses the mark) · "more than one polypeptide" (quaternary) · "hydrophilic R groups on the outside so soluble"
Bond name: "**peptide**". **Not accepted**: "dipeptide bond". (Avoid "amide": accepted once, rejected later.)
Drawing an amino acid: central C **with its H**, amine, carboxyl and R group. Check every C has 4 bonds, N 3, O 2, H 1.
Collagen: "fibrous", "**three polypeptide chains** / triple helix", "hydrogen bonds **between the chains**", "every third amino acid is glycine" → tight packing, "cross-links" → tensile strength.
Heating: "**hydrogen bonds break**", R groups vibrate more; "**peptide bonds are not affected**", so the primary structure is unchanged. A thermostable protein has "more / stronger bonds", so "more energy is needed to break them".
Often asked as: draw an amino acid or a peptide bond; describe collagen's structure; explain why a changed amino acid stops a protein working; explain thermostability.

BEYOND THE SPEC, BUT EXAMINED

- **Relative bond strengths**: disulfide (covalent) > ionic > hydrogen. More disulfide or ionic bonds → more energy needed to break the tertiary structure → more **thermostable** (WBI11 Jun 2025).
- **Sickle cell haemoglobin (HbS)**: valine's non-polar R group turns away from water and forms **hydrophobic interactions** with other haemoglobin molecules, so they stick together in the cytoplasm of red blood cells (WBI11 Jan 2022).
- **Collagen mutations**: replacing a glycine with an amino acid with a larger R group stops tight packing and weakens the triple helix (osteogenesis imperfecta) (WBI11 Jan 2025).
- **Proinsulin → insulin**: a protease hydrolyses peptide bonds to cut out a section of the chain (WBI11 Oct 2022).
- Proteins contain **C, H, O, N** and often **S** (WBI11 Oct 2025).

COMMON MISTAKE

"**Dipeptide bond**" is the commonest error. It's a **peptide** bond.
Leaving the H off the central carbon when drawing an amino acid.
Not saying where hydrogen bonds are: **between the chains** (collagen) or **between amino acids** (secondary structure).
Writing "iron" for haemoglobin: say "**iron ion (Fe^{2+})** in a haem group".

RECOMMENDED PRACTICAL: BIURET TEST FOR PROTEIN

Add biuret reagent (copper(II) sulfate in alkaline solution) to the sample. **Purple/lilac** = protein (it reacts with peptide bonds); stays **blue** = no protein. Using colour standards of known protein concentration makes it semi-quantitative, as in CP1.

SPEC 2.7 Enzymes: how they work

WHAT THIS SPEC POINT ASKS

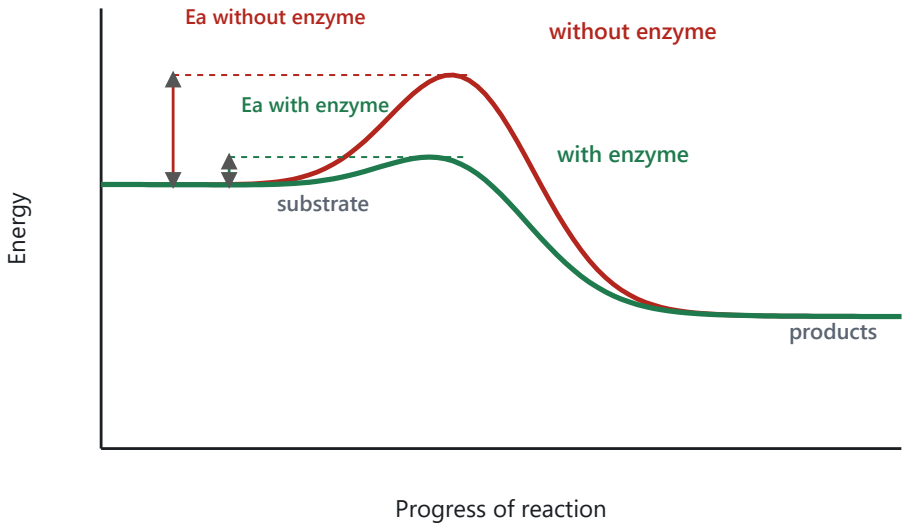
(i) Understand how enzymes work, and their specificity, in terms of their 3-D structure. (ii) Understand that enzymes are biological catalysts that lower activation energy. (iii) Know that intracellular enzymes work inside cells and extracellular enzymes outside them.

2.7 Mechanism and specificity

- Enzymes are **globular proteins**. Their tertiary structure creates an **active site** with a specific shape.
- Only a substrate with a **complementary** shape can fit and bind, forming an **enzyme–substrate complex**. That's why each enzyme is **specific** to one substrate (or one type of reaction).
- **Induced fit** model: the active site **changes shape slightly** as the substrate binds, moulding around it. This puts strain on the substrate's bonds (or brings reacting groups close together), making the reaction easier. Products are released and the enzyme is **unchanged**, ready to be reused.
- (The older **lock-and-key** model treated the active site as rigid. Induced fit replaced it because it explains why binding changes the enzyme's shape.)

2.7 Catalysts that lower activation energy

- **Activation energy** is the energy needed to start a reaction.
- Enzymes **lower the activation energy**, so reactions happen fast at body temperature, which would otherwise need far higher temperatures.
- They don't change the overall energy difference between substrate and product, and they aren't used up.



The enzyme lowers the activation energy (Ea). Starting and finishing energies are unchanged.

2.7 Where enzymes work

Intracellular (inside cells)	Extracellular (secreted, work outside cells)
e.g. **catalase** (breaks down toxic hydrogen peroxide), enzymes of respiration, DNA polymerase, RNA polymerase	e.g. **amylase** (starch → maltose), **trypsin** and **pepsin** (proteins), **lipase** (lipids) in digestion
made and used in the same cell	made in the cell, released by **exocytosis**

MARK-SCHEME LANGUAGE

Biological catalyst: define **both** words: "**protein** found in living organisms" + "**speeds up** a reaction **without being used up** / lowers the activation energy".
Mechanism: substrate "**binds to the active site**" → "**enzyme–substrate complex**" → activation energy lowered → bond broken by **hydrolysis**; name the bond (peptide, glycosidic). "Complementary" on its own isn't enough.
Specificity: the active site's shape depends on its **R groups**; the other substrate has a "**different shape / is not complementary**". Deal with **both** substrates.
Activation-energy diagrams: Ea is lower with the enzyme; the overall energy change is the **same**.
Often asked as: activation-energy diagram MCQs; why an enzyme acts on one substrate but not a similar one; enzyme action in a new context (lactase, proteases); inhibitors.

BEYOND THE SPEC, BUT EXAMINED

- **Competitive inhibitor** (WBI11 Oct 2025): shape **similar to the substrate**, so it **binds to the active site** → "**fewer enzyme–substrate complexes** formed per unit time". Adding more substrate outcompetes it, so the maximum rate can still be reached.
- **Non-competitive inhibitor** (WBI11 Oct 2025): binds **elsewhere on the enzyme**, at an **allosteric site** → "the **active site changes shape**" → fewer active sites the substrate can bind to. More substrate doesn't help: the maximum rate stays lower.
- **Warfarin** is a competitive inhibitor: its structure is similar to vitamin K, so it blocks the active site of the enzyme that recycles vitamin K for making clotting factors. Extra vitamin K competes with it. "Non-competitive" is not accepted (WBI11 Oct 2020).
- **Metabolic pathways**: the product of one enzyme is the substrate of the next (WBI11 Oct 2019).
- **Active-site size**: a larger active site can bind more substrate per unit time but may hold it less tightly (WBI11 Jun 2025).

COMMON MISTAKE

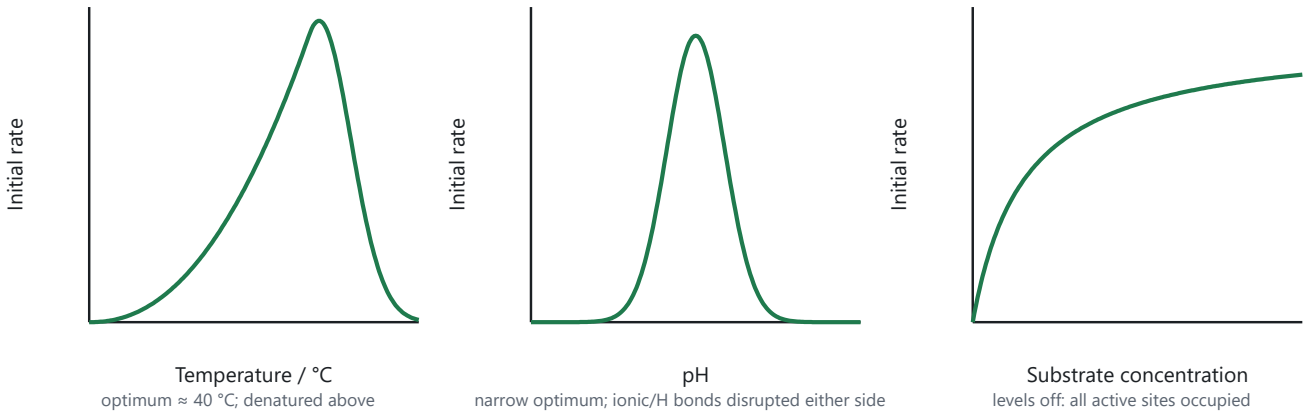
"The substrate and active site have the **same** shape." They have **complementary** shapes. And enzymes are **denatured**, not "killed".
Putting the active site on the **substrate**. Mixing up **lactose** (the substrate) and **lactase** (the enzyme).
Saying a protease breaks hydrogen or disulfide bonds: it hydrolyses **peptide** bonds.

SPEC 2.8

Core Practical 4: rate of enzyme-controlled reactions

WHAT THIS SPEC POINT ASKS

Investigate how temperature, pH, enzyme concentration and substrate concentration affect the initial rate of enzyme-catalysed reactions.



Typical shapes of the curves for the four factors (enzyme concentration gives a straight line while substrate is in excess).

2.8 How each factor works

- **Temperature**: as temperature rises, molecules have more **kinetic energy** → more frequent **successful collisions** between enzyme and substrate → more enzyme–substrate complexes → faster rate. Above the **optimum**, bonds holding the tertiary structure (especially hydrogen bonds) break → the active site changes shape → substrate no longer fits → enzyme **denatured** → rate falls fast. Denaturation is permanent.
- **pH**: the **H^+ / OH^-** concentration changes the **ionisation (charge) of R groups**, which affects the **ionic and hydrogen bonds** that hold the tertiary structure (and charges in the active site). Away from the optimum pH the active site changes shape; at extremes the enzyme denatures. Different enzymes have different optima (pepsin ≈ pH 2, amylase ≈ pH 7).
- **Enzyme concentration**: more enzyme → more active sites → rate rises **in proportion**, as long as substrate is in excess.
- **Substrate concentration**: rate rises at first (more substrate molecules → greater chance of **collisions with active sites**), then **levels off** when all active sites are **occupied** (saturated); enzyme concentration is now the limiting factor.

2.8 Measuring initial rate

- Rate is fastest at the start, because substrate is used up as the reaction goes on. So compare **initial rates**.
- Plot product formed (or substrate left) against time, draw a **tangent** to the curve at **t = 0**, and find its gradient.

WORKED EXAMPLE: INITIAL RATE FROM A TANGENT

O_2 produced by catalase is plotted against time. The tangent at t = 0 passes through (0 s, 0 cm^3) and (20 s, 9.0 cm^3).

Initial rate = change in y ÷ change in x = 9.0 ÷ 20 = **0.45 cm^3 s^{-1}**.

Rate can also be found as **1 ÷ time** to reach an end point (e.g. time for iodine colour to disappear), units s^{-1}.

CORE PRACTICAL 4: TWO STANDARD SET-UPS

Catalase + hydrogen peroxide (e.g. enzyme concentration): mix a **buffer**, H_2O_2 and catalase (or potato/yeast extract) in a conical flask connected to an upside-down measuring cylinder (or gas syringe) to collect O_2. Read the O_2 volume every 10 s for 60 s. Repeat for each enzyme concentration, 3 times each.

Amylase + starch (e.g. temperature or pH): mix starch and amylase (pre-incubated at the test temperature/pH in a water bath with a **buffer**). Every 10–30 s, put a drop onto **iodine** on a spotting tile;

record the time when it stays **orange-brown** (all starch digested). Or follow the fall in blue-black colour with a colorimeter.

Control variables: the other three factors (temperature with a **water bath**, pH with a **buffer**, and the volumes and concentrations), and the source and age of the enzyme. Leave solutions to reach temperature before mixing (**equilibrate**).

MARK-SCHEME LANGUAGE

Rising part of a rate–substrate curve: "more substrate molecules" → "greater chance of **collisions with the active site**" → more enzyme–substrate complexes. A bare "more collisions" is ignored.
Plateau: "**all active sites are filled** / enzyme is saturated". "Substrate is used up" is ignored.
High temperature: kinetic energy → "R groups vibrate" → bonds break → "active site distorted" → "enzyme–substrate complexes can't form". Write "**denatured**", not "starts to denature".
pH: "changes the **ionisation of R groups**" → bonds break → active site changes shape.
Rate = change in product (or substrate) **per unit time**, with units, e.g. $cm^3\ s^{-1}$ or $mmol\ dm^{-3}\ min^{-1}$. No time unit (or "a.u." alone) loses the mark.
Sketching the curve at the optimum: **steeper**, levels off **sooner**, at the **same** final level if the substrate is limiting.
Often asked as: explain each region of a rate–substrate curve; tangent for initial rate; conclusions from a pH × temperature table; immobilised v free enzyme data (6-mark).

BEYOND THE SPEC, BUT EXAMINED

- **Immobilised enzymes**, e.g. lactase trapped in **alginate** beads (WBI11 Jan 2026): can be **reused**; the product isn't contaminated with enzyme; active over a **wider range of pH and temperature** (higher optimum temperature), because "alginate holds the R groups in place / stabilises the active site", reducing vibration.
- **Rate equation** (given in the question): $V = V_{max} \times S \div (K + S)$, where K is the substrate concentration at $\frac{1}{2}V_{max}$ (WBI11 Jan 2020).
- Below the optimum, R groups vibrate a little and slightly distort the active site, but the extra collisions outweigh this, so the rate still rises (WBI11 Jun 2022).

COMMON MISTAKE

"At high temperature the enzyme moves too fast to bind." No: above the optimum the **active site changes shape** because bonds in the **tertiary structure** break, so the substrate is **no longer complementary**. Below the optimum, enzymes are **not** denatured; they just have less kinetic energy.
Explaining only one region of a curve: cover the rising part **and** the plateau. Say "molecules", not "particles".
Reading values off the curve when a **tangent** is needed; the tangent must touch the curve at the stated time. Give units and sensible decimal places.
"Immobilised enzymes are more efficient" or "the curve shifts right" score nothing: say what happens to activity and why.
Claiming an exact optimum from widely spaced readings. In a data table, draw conclusions rather than describing each row.

SPEC 2.9 Nucleotides, DNA and RNA

WHAT THIS SPEC POINT ASKS

(i) Know the structure of mononucleotides (deoxyribose or ribose + phosphate + a base: T, U, A, C or G) and of DNA and RNA as polynucleotides joined by condensation, forming phosphodiester bonds. (ii) Know how complementary base pairing and hydrogen bonding between two strands form the DNA double helix.

2.9 Nucleotides

- A **nucleotide** = a **pentose sugar** + a **phosphate group** + a **nitrogenous (organic) base**.
- The base is joined to **carbon 1** of the sugar, the phosphate to **carbon 5**.
- **Condensation** between the **phosphate** of one nucleotide and the **sugar** of the next (at carbon 3) forms a **phosphodiester bond**. A chain of these is a **polynucleotide**, with a **sugar–phosphate backbone**.

2.9 DNA vs RNA

	DNA	RNA
Sugar	**deoxyribose**	**ribose**
Bases	A, **T**, C, G	A, **U**, C, G
Strands	**two**, forming a **double helix**	**one**
Length	very long	much shorter
Job	stores genetic information stably	carries the code (mRNA), brings amino acids (tRNA), part of ribosomes (rRNA)

Bases are either **purines** (two rings: **A, G**) or **pyrimidines** (one ring: **C, T, U**).

2.9 Base pairing and the double helix

- The two DNA strands are held together by **hydrogen bonds** between **complementary** bases: **A–T** (2 hydrogen bonds) and **C–G** (3 hydrogen bonds). In RNA, A pairs with **U**.
- A purine always pairs with a pyrimidine, so the two strands stay a constant distance apart.
- The strands run in **opposite directions** (**antiparallel**) and twist into a **double helix**. The bases are on the inside, protected; the sugar–phosphate backbone is on the outside.
- Many hydrogen bonds together make DNA **stable**, yet each one is weak, so the strands can be **separated** for replication and transcription.

WORKED EXAMPLE: BASE PERCENTAGES

A DNA sample is 22% adenine. A pairs with T, so T = **22%**. A + T = 44%, so C + G = 56%, and C = G = **28%** each.

MARK-SCHEME LANGUAGE

DNA only: **deoxyribose, thymine**. RNA only: **ribose, uracil**. Both: **phosphate, adenine, cytosine, guanine**. Write base names **in full**.
"**Phosphodiester** bonds formed by **condensation** (loss of water)"; complementary bases held by **hydrogen** bonds in a "**double helix**" ("helix" alone loses the mark).
Purine pairs with pyrimidine "so the strands are parallel", i.e. the helix has a **constant width**.
mRNA v tRNA: **both** single-stranded, ribose, A/U/C/G, phosphodiester bonds; mRNA is **linear**, of variable length, with **codons**; tRNA is **clover-leaf** shaped with **hydrogen bonds**, an **anticodon** and an **amino acid binding site**. "tRNA carries an amino acid" is ignored as a *structural* difference.
Bonds in double-stranded RNA: phosphodiester (sugar–phosphate), covalent (base–sugar), hydrogen (between bases): one mark each.
Often asked as: DNA / RNA Venn diagrams; draw a nucleotide, or two joined; compare mRNA and tRNA; base percentage calculations.

BEYOND THE SPEC, BUT EXAMINED

- Nucleotides contain **C, H, O, N and P** (WBI11 Jun 2021).
- One turn of the helix is **3.4 nm** long and holds about **10 base pairs**, so adjacent base pairs are about **0.34 nm** apart (WBI11 Jan 2019).
- DNA mass and copies: after n divisions one molecule gives $\mathbf{2^n}$ copies; convert daltons to grams with the value given (1 Da = 1.67×10^{-24} g) (WBI11 Oct 2022).

COMMON MISTAKE

Phosphodiester bonds run **along** a strand; **hydrogen** bonds run **between** the strands. Don't mix them up when labelling diagrams, and don't join nucleotides in a strand by hydrogen bonds between their bases.
Drawing the base on the wrong carbon (it's C1), writing "P" for phosphate, or giving letters instead of base names.
Drawing a base pair when one nucleotide was asked for. In comparisons, write "**both**" for shared features.

SPEC 2.10 DNA replication and Meselson and Stahl

WHAT THIS SPEC POINT ASKS

(i) Understand DNA replication, including the role of DNA polymerase. (ii) Understand how Meselson and Stahl's experiment supported semi-conservative replication and refuted the competing theories.

2.10 Semi-conservative replication

1. The enzyme **DNA helicase** breaks the **hydrogen bonds** between bases and unwinds the double helix, separating the two strands.
2. **Each strand acts as a template.**
3. **Free DNA nucleotides** in the nucleus line up against the exposed bases by **complementary base pairing** (A–T, C–G), held by hydrogen bonds.
4. **DNA polymerase** joins adjacent nucleotides by **condensation reactions**, forming **phosphodiester bonds** in the new strand. It also **proofreads**, checking each new base pair and correcting mismatches.
5. Each new DNA molecule contains **one original strand and one new strand**: **semi-conservative** replication.

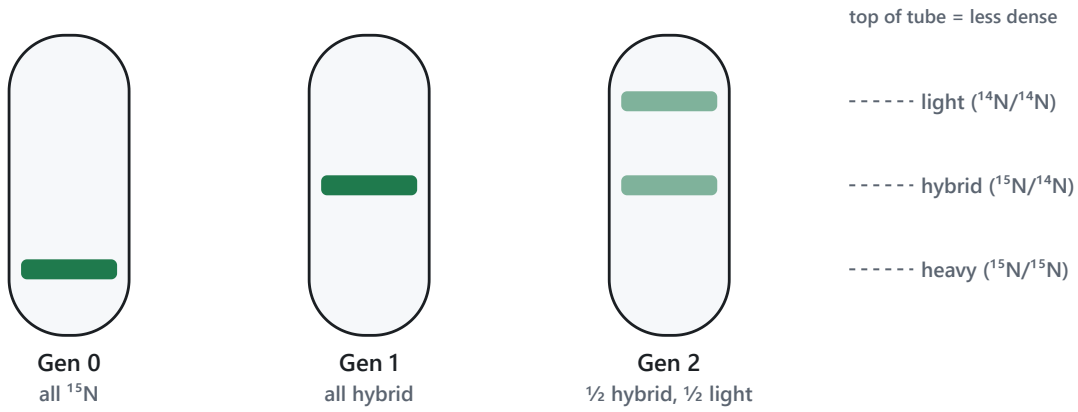
Replication happens before cell division, so each daughter cell gets an identical copy of the DNA.

2.10 Three competing theories

- **Conservative**: the original double helix stays intact; the copy is made entirely of new DNA.
- **Semi-conservative**: each new molecule is one old strand + one new strand.
- **Dispersive**: each new molecule is a patchwork of old and new sections along both strands.

2.10 Meselson and Stahl's experiment

1. Grew bacteria for many generations on a medium containing **heavy nitrogen (^{15}N)**, so all their DNA was "heavy" (nitrogen is in the bases).
2. Moved them to **light nitrogen (^{14}N)** and let them divide once, then twice.
3. Extracted the DNA each time and **centrifuged** it; heavier DNA settles **lower** in the tube.



Results: heavy band at the start; one intermediate (hybrid) band after one generation; one hybrid and one light band after two generations.

Generation in ^{14}N	Conservative predicts	Dispersive predicts	Semi-conservative predicts	Observed
1	two bands: heavy + light	one intermediate band	one intermediate band	**one intermediate band**
2	heavy + light (more light)	**one** band (between intermediate and light)	**intermediate + light** (equal amounts)	**intermediate + light**

- **Generation 1** rules out **conservative** (no heavy band).
- **Generation 2** rules out **dispersive** (two bands, not one).
- Only **semi-conservative** fits both.

EXAM TIP

For each model, state **what it predicts** and **why the result rules it out**. Explain the intermediate band: "each molecule has one ^{15}N strand and one ^{14}N strand, so it has an intermediate density."

MARK-SCHEME LANGUAGE

"**Semi-conservative**": "**both** parent strands act as **templates**"; each new molecule has "**one parent strand and one new strand**". Explain both halves of the word.
DNA polymerase: lines up nucleotides along the template and "forms **phosphodiester bonds**". "Forms hydrogen bonds" is ignored. If asked for roles (plural), give more than one.
Enzymes: **helicase**, **DNA polymerase**, (DNA ligase). "RNA polymerase" is wrong here.
Meselson–Stahl as % of DNA (heavy / hybrid / light): start 100 / 0 / 0 → gen 1: 0 / 100 / 0 → gen 2: 0 / 50 / 50 → gen 3: 0 / 25 / 75. Conservative after one generation: heavy + light; dispersive: one intermediate band.
Replication gives "**genetically identical**" copies, so daughter cells make the same proteins. "Identical" alone isn't enough.
Often asked as: explain semi-conservative replication; roles of DNA polymerase; predict the bands for each model; complete a percentage table; read density-peak graphs.

BEYOND THE SPEC, BUT EXAMINED

- Calculations of the **proportion of DNA** in each band after *n* generations: after 3 generations in ^{14}N, 2 of 8 molecules are hybrid and 6 are light → **25% hybrid, 75% light**. The number of hybrid molecules stays at 2 per original molecule.
- **DNA polymerase proofreading** keeps the mutation rate low (WBI11 Jan 2025).
- **DNA ligase** joins sections of new DNA together (WBI11 Jun 2019).
- **Replication bubbles** (WBI11 Jan 2022): replication starts at many points and runs in both directions. Copying 1.5×10^8 bases from one point at 50 bases s^{-1} would take $1.5 \times 10^8 \div 50 = 3 \times 10^6$ s ≈ **833 h**; with many origins it takes about 10 h.
- **Density graphs**: ^{15}N, hybrid and ^{14}N DNA give peaks at different densities, and peak height shows the **amount** of DNA (WBI11 Jan 2025).

COMMON MISTAKE

Mixing up "strand" and "molecule": each new **molecule** keeps one old **strand**.
Careless band drawings: put bands on the heavy, hybrid and light positions and show relative amounts by width. Fill every table cell (write 0, don't leave it blank).

SPEC 2.11

The nature of the genetic code

WHAT THIS SPEC POINT ASKS

Understand the nature of the genetic code: triplet, non-overlapping and degenerate.

- **Triplet**: a sequence of **three bases** (a **codon** on mRNA) codes for **one amino acid**. $4^3 = 64$ possible triplets, which is plenty for 20 amino acids (a doublet code would give only $4^2 = 16$).
- **Non-overlapping**: each base is read **once**, as part of **only one** triplet.
- **Degenerate**: most amino acids are coded for by **more than one** triplet (e.g. glycine by GGU, GGC, GGA, GGG). This means some substitution mutations change no amino acid (2.14).
- Some triplets are **start** (AUG) or **stop** (UAA, UAG, UGA) signals.
- (The code is also **universal**: the same triplets code for the same amino acids in almost all organisms.)

MARK-SCHEME LANGUAGE

Non-overlapping: "a base in one triplet **does not code for any other** amino acid" · Degenerate: "**most amino acids are coded for by more than one codon**" (accept: only Met and Trp have one codon) · "a sequence of 12 bases codes for **4** amino acids, because each amino acid needs 3 bases" · Universal: "the **same triplet codes for the same amino acid in (almost) all organisms**"
Triplet: "**three bases code for one amino acid**", with an example from the table (e.g. AAA = Lys). Each feature needs a table example, and don't define a term with the term itself.
Why a triplet: "two bases would only give **16** (4^2) combinations, not enough for **20** amino acids" (plus start and stop); three give 64 (4^3).
Degenerate also "**minimises the effect of mutations**".
Stop codons: "**no more amino acids can be added**". 3 of the 64 codons are stop codons, so 61 ÷ 64 = **95.31%** code for amino acids.
Often asked as: describe each feature using a codon table (6-mark); why a triplet and not a doublet; count bases, codons or amino acids; explain why a substitution had no effect.

NON-OVERLAPPING VS OVERLAPPING

mRNA: A U G G C A U U C
Read **non-overlapping**: AUG · GCA · UUC → 3 codons → 3 amino acids.
If it overlapped, AUG, UGG, GGC, GCA… would all be read, so one base change would alter several amino acids.

BEYOND THE SPEC, BUT EXAMINED

- The start codon **AUG also codes for methionine**, so Met appears inside polypeptides as well as at the start (WBI11 Jun 2024).
- Reading a **codon wheel**: start at the centre with the first base and work outwards (WBI11 Oct 2025).

COMMON MISTAKE

Check whether a table is written as **mRNA codons** or **DNA triplets**. A DNA template triplet **CAG** gives the mRNA codon **GUC**. Always convert to mRNA first before using a codon table.
Swapping **degenerate** (several codons for one amino acid) and **non-overlapping** (each base read once). It's the **code** that is degenerate, not DNA or the bases.

SPEC 2.12

What a gene is

WHAT THIS SPEC POINT ASKS

Know that a gene is a sequence of bases on a DNA molecule that codes for a sequence of amino acids in a polypeptide chain.

- A **gene** is a **sequence of bases on DNA** that codes for the **sequence of amino acids** in a **polypeptide**.
- Its position on a chromosome is its **locus**. One DNA molecule (chromosome) carries many genes.
- The chain of cause and effect: **base sequence** → **amino acid sequence (primary structure)** → **3-D shape** → **function** of the protein.

COUNTING

A polypeptide of 150 amino acids needs at least 150 × 3 = **450 bases** in the coding part of the gene (plus 3 for the stop codon).

MARK-SCHEME LANGUAGE

Gene: "a **sequence of (DNA) bases** that codes for a **(poly)peptide**" (accept "primary structure").
Allele: "a **version / form of a gene**". Gene v allele questions need **named examples** from the context, or they score 1.
mRNA "carries a **copy of the gene** to the ribosomes". "mRNA copies DNA" or "a copy of DNA" is too vague.

COMMON MISTAKE

Mixing up gene and allele. The gene is the section of DNA for one polypeptide (e.g. the CFTR gene); alleles are its versions (e.g. the normal and the faulty CFTR allele).

SPEC 2.13

Protein synthesis: transcription and translation

WHAT THIS SPEC POINT ASKS

(i) Understand transcription and translation, including RNA polymerase, mRNA, tRNA, ribosomes, and start and stop codons. (ii) Understand the roles of the DNA template (antisense) strand, codons on mRNA and anticodons on tRNA.

2.13 Transcription (in the nucleus)

1. The DNA double helix **unwinds** and **hydrogen bonds break** in the region of the gene, separating the strands.
2. Only **one strand**, the **template (antisense) strand**, is used.
3. **Free RNA nucleotides** pair with the exposed DNA bases by **complementary base pairing** (A–U, T–A, C–G, G–C).
4. **RNA polymerase** joins the RNA nucleotides with **phosphodiester bonds**, forming **mRNA**.
5. mRNA **detaches**; the DNA re-forms its helix. mRNA leaves the nucleus through a **nuclear pore**.

The mRNA has the **same sequence as the coding (sense) strand**, except **U replaces T**.

2.13 Translation (on ribosomes in the cytoplasm)

1. The mRNA attaches to a **ribosome**.
2. **tRNA** molecules each carry a **specific amino acid** and have an **anticodon** (three bases).
3. The tRNA whose anticodon is **complementary** to the **start codon** (AUG) binds first.
4. The next tRNA binds to the next codon. The ribosome holds **two tRNAs** at a time.
5. A **peptide bond** forms between the two amino acids (**condensation**), using ATP.
6. The ribosome **moves along one codon**; the first tRNA leaves (to collect another amino acid).
7. This repeats until a **stop codon**, which has no tRNA; the **polypeptide is released**.

2.13 Worked conversion (a very common question)

DNA → MRNA → TRNA → AMINO ACIDS

	Triplet 1	Triplet 2	Triplet 3
DNA template (antisense) strand	TAC	GGT	ACT
mRNA codon	AUG	CCA	UGA
tRNA anticodon	UAC	GGU	—
Amino acid	Met (start)	Pro	stop

The **coding (sense)** strand would read ATG CCA TGA, the same as mRNA with T for U.
Shortcut: the **anticodon** is the template triplet with U in place of T (TAC → UAC).

2.13 mRNA vs tRNA

	mRNA	tRNA
Shape	single, linear strand	single strand folded into a **clover-leaf**, held by H-bonds
Key feature	**codons**	**anticodon** at one end; **amino acid binding site** at the other
Job	carries the gene's code to the ribosome	brings the specific amino acid to the ribosome

MARK-SCHEME LANGUAGE

Exact names: "**transcription**", "**translation**", "**RNA polymerase**" (not DNA polymerase, helicase or just "polymerase"). Hedging between two names scores 0.
mRNA: "carries a copy of the gene out of the nucleus to the ribosomes"; its codons set the order of the amino acids.
tRNA: "carries a **specific** amino acid"; anticodon "**complementary** to the codon"; "holds the amino acid in place until the peptide bond forms".
Ribosome: site of translation; "**holds (two) tRNAs** on the mRNA" while the peptide bond forms.
Start codon: "starts translation" and codes for **methionine**. Stop codons have **no tRNA**.
Effect of a mutation: "the change is **copied into the mRNA**" → a different **codon** → "a tRNA carrying a different amino acid binds" → it's joined into the polypeptide by a peptide bond.
Often asked as: DNA → mRNA → anticodon → amino acids from a table; roles of mRNA, tRNA, RNA polymerase or ribosomes; compare start and stop codons.

BEYOND THE SPEC, BUT EXAMINED

- **RNA polymerase binds to the promoter** (a base sequence just before the gene), separates the strands and forms phosphodiester bonds (WBI11 Jun 2024).
- **Start v stop codons**: 1 start codon (AUG, methionine) and 3 stop codons (UAA, UAG, UGA). No tRNA has an anticodon for a stop codon, so the polypeptide is released (WBI11 Jun 2024).
- **mRNA medicines**: lipid nanoparticles protect the mRNA from hydrolytic enzymes and fuse with the cell surface membrane; the mRNA is then translated (WBI11 Oct 2019).
- **RNA interference and antisense drugs** stop the mRNA being **translated**; transcription carries on (WBI11 Jan 2025).
- **Codominance** at the molecular level: both alleles are transcribed and translated, so both proteins are made (WBI11 Jan 2023).

COMMON MISTAKE

Mixing up **DNA polymerase** (replication) with **RNA polymerase** (transcription). Writing **T** in mRNA or tRNA (it's **U**). The **anticodon is on tRNA**, the **codon on mRNA**.
Saying peptide bonds form in transcription: they form in **translation**.
Thinking stop codons have a tRNA. Giving the mRNA sequence when the question asks for the amino acids.
Naming an enzyme or molecule when the question asks for its **role**.

SPEC 2.14

Mutations

WHAT THIS SPEC POINT ASKS

(i) Understand how errors in DNA replication cause mutations (substitution, insertion, deletion). (ii) Know that some mutations cause cancer or genetic disorders, but many have no observable effect.

A **gene mutation** is a change in the **base sequence** of DNA, usually from an error during **replication**.

Type	What happens	Effect
Substitution	one base replaced by another	changes **one codon** only: may give the **same** amino acid (degenerate code: *silent*), **one different** amino acid, or a **stop codon** (shortened protein)
Insertion	an extra base added	**frameshift**: every codon after it is read differently → many amino acids change; often a stop codon appears
Deletion	a base lost	**frameshift**, as for insertion

WORKED EXAMPLE

Original mRNA: AUG **G**CA UUC → Met – Ala – Phe
Substitution (G→C): AUG **C**CA UUC → Met – **Pro** – Phe (one amino acid changes)
Deletion of that G: AUG CAU UC… → Met – **His** – … (everything after the mutation changes)

2.14 Why a mutation may or may not matter

- **No effect**: the new codon codes for the **same** amino acid; or the change is in non-coding DNA; or the new amino acid has similar properties and doesn't change the shape.

- **Harmful**: a different amino acid changes the **primary structure** → different bonds between R groups → different **tertiary structure** → the protein doesn't function (e.g. an enzyme's active site no longer complementary). Examples: **cystic fibrosis** (2.16), sickle cell anaemia.
- **Cancer**: mutations in genes that control **cell division** can lead to uncontrolled division and a **tumour**.
- Mutations in **gametes** can be **inherited**; mutations in body cells are not.

MARK-SCHEME LANGUAGE

Mutation: "a change in the **base sequence** of DNA".
Use the spec's names: **substitution, insertion, deletion**. "Point", "frameshift", "addition", "nonsense" and "duplication" are **ignored** as type names.
Substitution: affects **one triplet**; may be **silent** (degenerate code, or a non-coding region); may change **one amino acid** → different bonds between R groups → different structure and activity; may make a **stop codon** → shorter polypeptide, or none.
Insertion / deletion: shifts the reading frame, so **every later amino acid** may change; the **earlier** in the gene, the bigger the effect.
Cancer: "cells **dividing out of control**" as a result of mutation; they may spread.
Often asked as: name the mutation type from two sequences; use a codon table to explain the effect on the protein (6-mark); explain why a mutation had no effect.

BEYOND THE SPEC, BUT EXAMINED

- **Chromosome mutations** appear in school assessments and some data questions: whole chromosomes gained or lost through **non-disjunction** in meiosis (e.g. **Down's syndrome**, trisomy 21: three copies of chromosome 21), and changes to chromosome structure (deletion, duplication, inversion, translocation of a section). These change many genes at once.
- **Why mutation rates differ** (WBI11 Oct 2024): faster cell division (more replication), less **DNA repair or proofreading**, more exposure to **mutagens** such as UV, more DNA. Mutations also **build up with age**.
- **Proto-oncogenes**: normal genes that control cell division; mutated, they can cause cancer (WBI11 Jan 2025).

COMMON MISTAKE

"Substitution of bases" just repeats the name: say **which** base changed and what that does to the codon and the amino acid.
Generic answers: apply the mutation to the **table or sequence given** and link it to the protein's **function**.

EXAM TIP

Explain the chain fully: base sequence → **codon/triplet** → **amino acid sequence (primary structure)** → **bonds / 3-D (tertiary) shape** → **function**.

SPEC 2.15 Inheritance: terms, crosses, pedigrees, sex linkage

WHAT THIS SPEC POINT ASKS

(i) Understand gene, allele, genotype, phenotype, recessive, dominant, codominance, homozygote, heterozygote. (ii) Understand monohybrid inheritance, including interpreting pedigree diagrams. (iii) Understand sex linkage on the X chromosome, including red-green colour blindness.

2.15 Key terms

Term	Meaning
Gene	a sequence of bases on DNA coding for a polypeptide; found at a **locus**
Allele	a different **version** of the same gene
Genotype	the **alleles** an organism has (e.g. Ff)
Phenotype	the **observable characteristics**, from genotype + environment
Dominant	allele always expressed when present (capital letter)
Recessive	allele expressed only when **no dominant allele** is present (homozygous)
Codominance	**both** alleles expressed in the heterozygote (e.g. roan cattle, blood group AB)
Homozygote	two **identical** alleles (FF or ff)
Heterozygote	two **different** alleles (Ff); if one is a harmful recessive, the person is a **carrier**

2.15 Monohybrid crosses

TWO HETEROZYGOUS PARENTS (DOMINANT / RECESSIVE)

Parents: **Ff × Ff** · Gametes: F, f and F, f

	F	**f**
F	FF	Ff
f	Ff	ff

Genotypes 1 FF : 2 Ff : 1 ff · Phenotypes **3 dominant : 1 recessive**. Probability of an affected (ff) child = **¼ (25%)**, for *each* pregnancy.

CODOMINANCE

Red cattle (C^RC^R) × white (C^WC^W) → all **roan** (C^RC^W). Roan × roan → **1 red : 2 roan : 1 white**.

Always lay out: **parental phenotypes → genotypes → gametes → Punnett square → offspring genotypes → phenotypes and ratio / probability**. Choose letters whose capitals look different from the lower case (F/f, not S/s or C/c).

2.15 Reading pedigree diagrams

- Squares = male, circles = female; shaded = affected; horizontal line = partners; vertical line down to children; generations numbered I, II, III.
- **Two unaffected parents with an affected child → the condition is recessive**, and both parents are **carriers** (heterozygous).
- Two affected parents with an unaffected child → dominant.
- **Start with the affected (homozygous recessive) people**, whose genotypes you know for certain, and work outwards.
- An unaffected child of two carriers has a **2/3** chance of being a carrier (FF : Ff = 1 : 2 among the unaffected).

2.15 Sex linkage

- Sex is determined by the sex chromosomes: female **XX**, male **XY**. The **Y is much smaller** and lacks many genes found on the X.
- A **sex-linked** gene is carried on the **X chromosome**. Males have only **one X**, so a single recessive allele is **expressed** in a male. Males can't be carriers.
- Fathers pass their X only to **daughters**, so a son can't inherit an X-linked condition from his father.
- **Red-green colour blindness**: recessive allele $\mathbf{X^b}$; normal $\mathbf{X^B}$.

CARRIER MOTHER × NORMAL FATHER

Parents: $X^BX^b \times X^BY$ · Gametes: X^B, X^b and X^B, Y

	$\mathbf{X^B}$	**Y**
$\mathbf{X^B}$	X^BX^B normal girl	X^BY normal boy
$\mathbf{X^b}$	X^BX^b carrier girl	X^bY **colour-blind boy**

Probability of a colour-blind child = ¼. Probability a **son** is colour-blind = ½. No daughters are affected (but half are carriers).

BEYOND THE SPEC, BUT EXAMINED

- **ABO blood groups**: three alleles (I^A, I^B, I^O). I^A and I^B are **codominant**; I^O is recessive. I^AI^B = group AB; I^OI^O = group O. Used in "which baby belongs to which parents" questions.
- **Test cross**: cross an individual with the dominant phenotype with a **homozygous recessive**. All dominant offspring → it was homozygous; about half recessive offspring → it was heterozygous.
- **Haemophilia** is another **X-linked recessive** condition (a missing clotting factor, so blood doesn't clot properly: link to 1.11). It is set up and solved exactly like colour blindness, and explains why far more males are affected.
- Recessive alleles often *are* expressed (they make a non-functional protein), but one dominant allele makes enough working protein, so the phenotype is normal.
- **ZW sex determination** (birds, Komodo dragons): the **female** is ZW (heterogametic) and the male ZZ (homogametic), the reverse of humans (WBI11 Jan 2026).
- **Chi-squared test** to see if offspring numbers fit an expected ratio: work out $(O - E)^2/E$ for each class, add them up, and compare with the critical value at p = 0.05 (WBI11 Oct 2021).

MARK-SCHEME LANGUAGE

Genotype: "the **combination of alleles**" (not "genes"). Phenotype: "**observable characteristic**".
Codominance: "**both alleles are expressed**" in the heterozygote, whose phenotype differs from both homozygotes. Not "both dominant".
Genetic diagram marks: parental genotypes → gametes → offspring genotypes → **phenotypes linked to genotypes** → ratio or number. The phenotype mark is the one most often lost. Label ratios ("3 orange : 1 white"), and give **numbers** if numbers are asked for: 284 offspring in a 1 : 2 : 1 ratio → 71 : 142 : 71.
Sex linkage: allele "**on the X chromosome**", recessive; "males need only **one** copy, females need **two**"; "an affected father can't pass it to his sons, because they get his **Y**".
Pedigrees: one reasoned sentence per individual, e.g. "**dominant** because 1 and 2 both have the trait but have a child without it"; "**not X-linked** because female 6 shows the recessive trait but her father doesn't".
Probability of an affected child **of a given sex** = probability affected × ½, e.g. Ff × Ff → affected boy = ¼ × ½ = **⅛**.
Often asked as: genetic diagrams; probability MCQs; prove from a pedigree that a condition is dominant / recessive, or not X-linked; sex-linked crosses in new contexts (haemophilia, fruit flies).

COMMON MISTAKE

Writing sex-linked genotypes without the X and Y (always write X^B, X^b, Y), or giving the Y an allele. Also read the question carefully: "probability that a *child* is affected" differs from "probability that a *boy* is affected".
"Recessive **gene**" or "recessive **disease**": it's a recessive **allele**.
Using X and Y symbols for an **autosomal** cross: it scores 0 or is capped.
Giving the genotype ratio when the phenotype ratio was asked for, or an unlabelled "1 : 3". Forgetting that females can be affected by X-linked conditions.

SPEC 2.16

Cystic fibrosis

WHAT THIS SPEC POINT ASKS

Understand how a gene mutation in cystic fibrosis affects the gas exchange, digestive and reproductive systems.

2.16 The cause

- CF is caused by a **recessive allele** of the **CFTR gene**. CFTR is a **chloride ion channel** protein in the membranes of epithelial cells.
- The commonest mutation, **ΔF508**, is a **deletion of three bases** (one codon), so one amino acid (**phenylalanine**) is missing. The protein **folds wrongly** and doesn't work, or isn't inserted in the membrane.

2.16 From faulty protein to sticky mucus (learn this chain)

1. Normally, CFTR lets **Cl^- ions** move **out** of epithelial cells into the mucus; Na^+ follows; this **lowers the water potential** of the mucus, so **water moves out** of the cells **by osmosis**, keeping the mucus runny.
2. In CF, Cl^- isn't moved out of the cells, and **Na^+ ions are absorbed** into the cells from the mucus. The water potential **inside the cells** falls below that of the mucus, so **water moves out of the mucus into the cells by osmosis** → the mucus is **dehydrated**: **thick and sticky**.

2.16 Effects on the three systems

System	What goes wrong	Consequence
Gas exchange	sticky mucus blocks **bronchioles**; **cilia can't move** it	fewer alveoli ventilated → **smaller surface area** and **longer diffusion distance** → less O_2 uptake (breathlessness); bacteria trapped in mucus → frequent **lung infections**; coughing
Digestive	mucus blocks the **pancreatic duct**; mucus coats the small intestine lining	**digestive enzymes can't reach the small intestine** → food not fully digested; pancreas damaged (cysts, possibly diabetes); thick mucus reduces **absorption** → malnutrition, difficulty gaining weight
Reproductive	thick mucus blocks the **sperm ducts (vas deferens)** in men, or they never form; thick **cervical mucus** in women, and mucus may block the oviducts	men usually **infertile**; women's fertility reduced (sperm can't pass through the cervix to reach the egg)

Inheritance: two carrier parents (Ff × Ff) → **1 in 4** chance each child has CF.

EXAM TIP

Examiners reward the **full chain**: Cl^- not transported → water potential → osmosis → thick mucus → *then* the effect on the specific system. Link gas exchange answers back to **Fick's law** (2.1).

MARK-SCHEME LANGUAGE

The chain: "mutation in the CFTR **gene**" → "CFTR (channel) protein not functioning" → "**chloride ions do not move out** of the cells" → Na^+ moves in → water potential inside the cells falls → "**water leaves the mucus**" by osmosis → "mucus is **dehydrated**", thick and sticky. "Water concentration" is ignored.
Lungs: "**cilia cannot move** the mucus away"; mucus "**traps bacteria**" → infections, coughing, tissue damage; blocked airways → lower concentration gradient, smaller surface area, longer distance → **slower diffusion**.
Digestion: mucus **blocks the pancreatic duct** → enzymes don't reach the small intestine → food not fully digested or absorbed.
Reproduction: "mucus **prevents sperm passing through the cervix**"; mucus may block the oviduct.
ΔF508: "**deletion** removing the **three bases / codon** that code for phenylalanine". **Not accepted**: insertion, or a stop codon.
Often asked as: explain why the mucus is sticky; explain the effect on one system; interpret drug-trial data (FEV1); CF probabilities from carrier parents.

BEYOND THE SPEC, BUT EXAMINED

- **Sweat test**: CFTR also moves Cl^- in sweat glands, so people with CF have salty sweat. Different CFTR mutations leave different amounts of working protein, so chloride levels and symptoms vary (WBI11 Jun 2021).
- **Drug trials**: lung function is measured as **FEV1** (volume breathed out in the first second) with a spirometer; compare the **change in mean** FEV1 (WBI11 Jan 2023).
- A carrier man's sperm: meiosis shares the alleles equally, so **50%** of his sperm carry the faulty allele (WBI11 Oct 2025).

COMMON MISTAKE

"The CFTR **protein** is mutated." The **gene** is mutated; the protein it codes for is faulty.
Writing a generic CF essay: use the diagram or data in the question.

SPEC 2.17 Genetic screening

WHAT THIS SPEC POINT ASKS

(i) Understand the uses of genetic screening: identifying carriers, pre-implantation genetic diagnosis (PGD), and prenatal testing, including amniocentesis and chorionic villus sampling. (ii) Understand the implications of prenatal genetic screening.

2.17 Uses

- **Carrier screening**: tests adults (blood or cheek cells) with a family history of a recessive condition, to see if they carry the allele. Couples can make informed decisions about having children.
- **Pre-implantation genetic diagnosis (PGD)**: during **IVF**, one or two cells are removed from each early embryo and tested; only embryos **without** the allele are implanted. Avoids a later termination, but is expensive and unaffected embryos may be discarded.
- **Prenatal testing**: tests the fetus during pregnancy.

2.17 Amniocentesis vs CVS

	Amniocentesis	Chorionic villus sampling (CVS)
Sample	**amniotic fluid** containing fetal cells	cells from the **chorionic villi** (fetal part of the placenta)
How	needle through the abdomen, guided by **ultrasound**	needle through abdomen, or tube through the cervix, with ultrasound
When	about **15–20 weeks**	earlier: about **10–14 weeks**
Miscarriage risk	about **0.5–1%**	about **1–2%**
Results	slower (cells may be cultured: 2–3 weeks)	faster

CVS gives an **earlier** result, so any decision (e.g. about termination) is made earlier, but carries a slightly **higher** risk. Results can occasionally be wrong (**false positives/negatives**). (Non-invasive testing of fetal DNA in the mother's blood avoids miscarriage risk.)

2.17 Implications of prenatal screening

- **Positive**: parents can prepare (medically, emotionally, financially), choose whether to continue the pregnancy, and the child can be treated early.
- **Negative**: risk of **miscarriage** from the procedure; difficult decisions about **termination**; false results; anxiety; pressure from others.

MARK-SCHEME LANGUAGE

Carrier screening: identifies **heterozygous** couples so they can make an informed decision, but "informed choice" alone isn't enough: **give the options**: no children or adoption; IVF with PGD so "only healthy embryos are implanted"; prenatal testing, then possibly termination. Result: fewer homozygous births.
Amniocentesis: "**amniotic fluid** from around the **fetus**" → fetal cells "analysed for the presence of the **faulty allele**". CVS: "cells / tissue from the **placenta**" ("chronic" villus is rejected).
Risks: "**increases the risk of miscarriage**" (name the procedure); false positives / negatives.
Reasons for a prenatal test: family history, to prepare, to choose an unaffected embryo. Abortion is **not accepted** as the reason.
Often asked as: explain how carrier screening reduces CF births; describe amniocentesis or CVS; 6-mark questions weighing up the options for a named disorder.

BEYOND THE SPEC, BUT EXAMINED

- **Other prenatal tests and timings** (WBI11 Jan 2025): **NIPT** (fetal DNA in the mother's blood, no miscarriage risk) from about 7–10 weeks; CVS; amniocentesis 15–20 weeks; ultrasound; **cordocentesis** (fetal blood from the umbilical cord) at about 22–24 weeks.
- PGD can also screen embryos for the **wrong number of chromosomes** (aneuploidy); it may give false hope and wastes embryos (WBI11 Oct 2021).

COMMON MISTAKE

Treating carrier screening (of adults) as prenatal screening, or confusing PGD (embryos in IVF) with CVS / amniocentesis (a fetus in the uterus).
Use the right word: **embryo** for PGD, **fetus** for CVS and amniocentesis, never "baby".
In 6-mark answers, take each option in turn, use the data given, and tie it to the named disorder. Offering gene therapy is not screening.

SPEC 2.18 Ethical and social issues of genetic screening

WHAT THIS SPEC POINT ASKS

Identify and discuss the ethical and social issues of genetic screening from a range of ethical viewpoints, including religious, moral and social ones.

2.18 Frameworks to use

- **Rights-based / deontological**: some actions are right or wrong in themselves (e.g. "every embryo has a right to life", or "people have a right to know or not to know").
- **Utilitarian**: the right choice gives the **greatest good for the greatest number**, weighing suffering prevented against harms.
- **Autonomy**: individuals should make informed decisions about their own bodies and families.

2.18 Arguments

For screening	Against / concerns
prevents suffering from a serious genetic disorder (utilitarian)	**religious** belief that life begins at conception: termination and discarding embryos are wrong
gives parents information and **choice** (autonomy)	miscarriage risk from amniocentesis/CVS could kill a healthy fetus
lets parents prepare; early treatment	could lead to "**designer babies**" (selecting for sex or non-medical traits)
reduces long-term care costs for health services	**discrimination**: implies that disabled lives are less valuable; insurers or employers might use results
PGD avoids late termination	results can be wrong (false positives/negatives); anxiety; the right **not** to know
	who decides which conditions are "serious enough"?

Genetic counselling before and after testing explains the risks, the meaning of results and the options, without directing the decision.

EXAM TIP

Balanced answers score highest: give **both sides**, name the **viewpoint** each argument comes from (religious, moral, social, economic), and relate it to the **specific technique** in the question (e.g. PGD discards embryos; amniocentesis risks miscarriage).

MARK-SCHEME LANGUAGE

Credited: termination is "**taking a human life**", which is unethical or against religious / cultural beliefs; the embryo "has a **right to life**" and "**cannot give consent**"; "**spare embryos** from IVF are destroyed"; relatives who share the genes "may be **exposed to unwanted facts**"; **false** positives / negatives; miscarriage risk **tied to CVS or amniocentesis**; the perceived worth of an affected child.
Positives: informed choice, time to prepare, fewer people affected.
Not credited: a bare "designer babies" (explain the selection of non-medical traits in IVF instead); "violation of privacy" (too vague).
Often asked as: discuss the ethics of a named test; ethical points inside 6-mark screening questions.

COMMON MISTAKE

One-sided answers: "discuss" means both sides.
Mentioning miscarriage without saying which procedure causes the risk.
Calling an IVF embryo a "fetus" or "baby".